Servo-pneumatic welding gun system, 2nd generation CPX-WELD-2G-SA

FESTO

System manual

Installation and commissioning

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[8136954]

Version 2020-06a

Designation CPX-WELD-2G-SA

Order no. 8136954

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Revisions index			
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Table 1.1 Revisions index

Specified directives/standards

Version status		
EN 418:1993-01		IEC 60204-1:2007-06
EN 14121-1:2007-12		IEC 60947-5-1:2000-08
EN 50178:1998-04		IEC 61131-2:2008-04
EN ISO 13849-1:2008-12		

Table 1.2 Directives/standards specified in the document

Notes regarding this documentation

This documentation describes the installation and commissioning process for the servo-pneumatic welding gun system CPX-WELD-2G-SA.

Identification of dangers and instructions on how to avoid them:

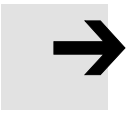


Warning
Dangers which can lead to death or serious injuries.

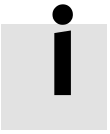


Caution
Dangers which could lead to minor injuries.

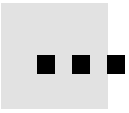
Other symbols:



Note
Damage to property or loss of function.



Recommendation, tip, reference to other documentation.



Necessary or useful accessories.



Information on environmentally friendly use.

Text symbols and markers:

- Activities that can be carried out in any order.

1. Activities which should be carried out in the specified sequence.

– General lists.

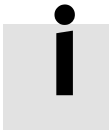
Notes regarding this description

This manual contains general basic information on the assembling, installing and commissioning of a controlled welding gun drive.

The documentation is only to be used by instructed and experienced persons. It does not contain the detail required of complete technical documentation for a regular mass-produced product.

Name	Contents
System manual “Servo-pneumatic welding gun system”	This description: Overview of the assembly, installation, commissioning and calibration of welding gun drives.
Brief instructions “Servo-pneumatic welding gun system”	Connection technology as well as information on assembly, installation and commissioning.
Software package WinSPAS	Commissioning software for welding gun drives.

Table 1.3 Documentation and software for the controlled welding gun system



Basic information regarding assembly and power supply of CPX terminals can be found in the CPX system description P.BE-CPX-SYS-...

For information on the CPX node used, see the corresponding description, e.g. P.BE-CPX-PNIO-DE for the CPX-M-FB34 with network protocol PROFINET IO.

Scope of delivery

Type code	Designation
CPX-WELD-2G-SA	Controller
SPSB-WELD-2G-SA	Control block: Proportional directional control valve MPYE, differential pressure regulator MPYD and manifold block
D-...-SPA-C-...-2G-SA	Cylinder
NEBU-M12G4-K-1.1-M12...4	Connecting cable
NEBU-M12G8-K-...-M12G8	Connecting cable

Table 1.4 Available components

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1. Safety and requirements for product use

1.1 Safety



Warning

The drives move dynamically and with great force.

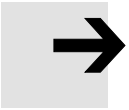
- Make sure that nobody can enter the sphere of influence of the drives and moving parts while the system is connected to energy sources.



Warning

Errors in installation or parameterisation can cause injury to people and material damage.

- Operate the system only if it has been professionally installed and parameterised.



Note

- A danger and risk analysis must be carried out for project planning of the servo-pneumatic welding gun system.
- Make sure the operator has the necessary level of training and is familiar with the details of intended use.

1.1.1 Intended use

The controller CPX-WELD-2G-SA and the associated components are exclusively intended for installation in a machine or an automated system. Depending on the bus node used, they can be connected to a fieldbus system and used to control welding guns.

It is imperative that the specified safety instructions and the intended use of the components is observed. The components and cables may only be used as follows:

- as intended in an industrial environment
- in an original state without unauthorised modifications. Only the conversions or modifications described in the documentation supplied with the product are permitted
- in a perfect technical condition

The limit values specified for pressures, temperatures, electrical data, torques etc. must be observed.

Observe the regulations of the trade associations, German Technical Control Board (TÜV), VDE stipulations or corresponding national laws and regulations.

1.1.2 Range of application and certifications

The product fulfils the requirements of EU directives and is marked with the CE marking.

Standards and test values, which the product observes and fulfils, can be found in the Technical data section. The product-relevant EU directives can be found in the declaration of conformity.

1.1.3 Target group

This manual is directed exclusively toward technicians trained in control and automation technology.

1.1.4 Service

Please consult your local Festo repair service if you have any technical problems.

1.2 Requirements for product use

- Comply with the specifications of the documentation. Follow all accompanying documentation and the documentation of any associated accessories.
- Take the following into consideration for the destination:
  - Applicable legal regulations
  - Regulations and standards
  - Regulations of the testing organisations and insurers
  - National specifications

For correct and safe use:

- Observe all warnings and notes.
- Comply with all load limits of the product and the connected components.

1.2.1 Qualification of specialized personnel

- The product may only be installed by specialized personnel with corresponding qualifications.

The following knowledge is required:

- installation and operation of electrical control systems
- applicable regulations for operating safety-engineering systems
- applicable regulations for accident prevention and occupational safety
- documentation and mode of operation of the product

1.3 Service

- Consult your local Festo service or write to the following e-mail address if you have any technical problems (➔ service_international@festo.com).

2. Component overview

2.1 System components

The Festo welding gun system is made up of the following electric and pneumatic devices.

- Controller
- Control block
- Main cylinder
- Compensating cylinder
- Various cables

2.1.1 Controller CPX-WELD-2G-SA

The controller takes over the following functions:

- Positioning and force control of the welding electrodes
- Control of the valve for gun compensation
- Fieldbus communication with the robotic control system
- Control of the additional pneumatic circuit and shutdown of the air supply

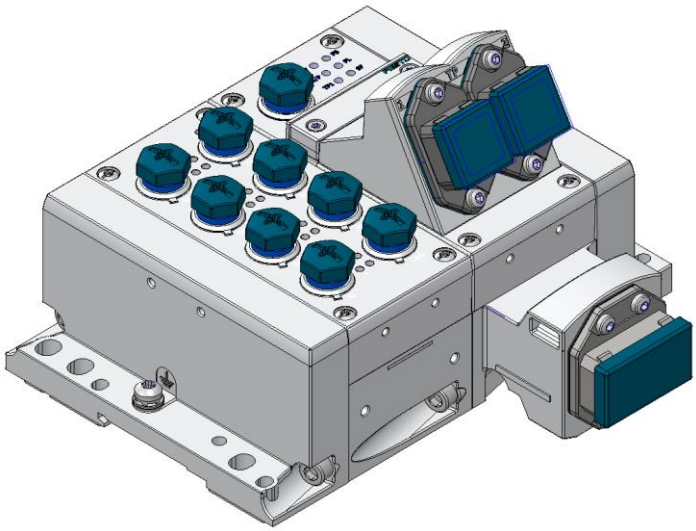


Figure 2-1 Controller CPX-WELD-2G-SA

2.1.2 Control block

The control block brings together the following:

- Proportional directional control valve MPYE for controlling the main cylinder
- Differential pressure regulator MPYD for controlling the compensating cylinder
- Manifold block with:
  - 3/2-way valve for the additional pneumatic circuit
  - 3/2-way valve for shutting down the air supply
  - Air filter (5 µm)

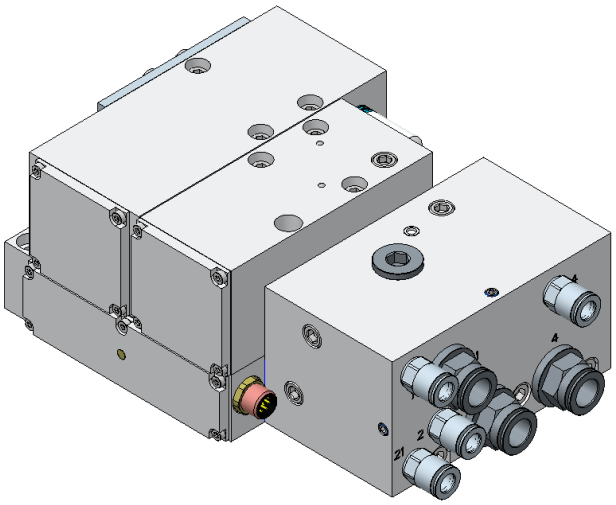


Figure 2-2 Control block with air supply shutdown

2.1.3 Main cylinder

Included in the main cylinder:
- Displacement sensor
- Pressure sensors
- Interface to controller CPX-WELD-2G-SA
- Exhausting of the main cylinder if the compressed air supply is removed
- Locking the main cylinder if the load voltage supply is switched off

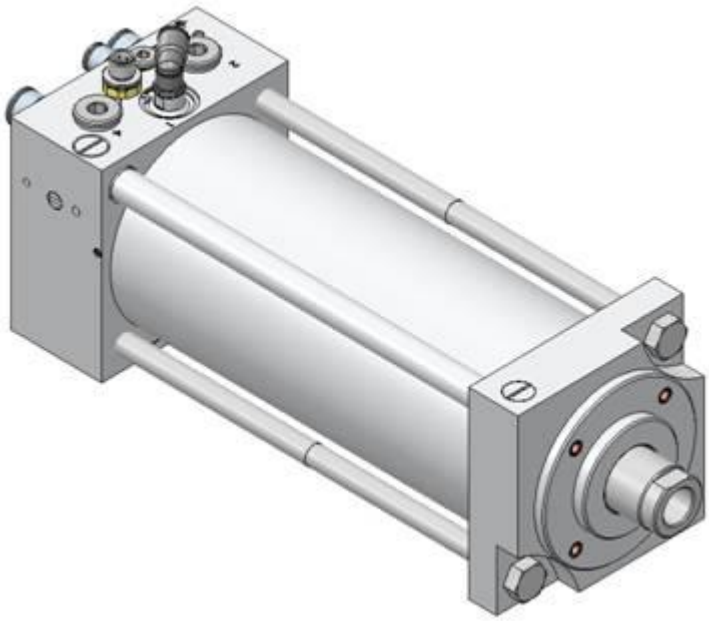


Figure 2-3 Main cylinder

Note
When the air supply shutdown function is activated, the chamber pressures in the cylinder are locked-in via the additional pneumatic circuit.

3. Mounting

The servo-pneumatic welding gun system is supplied from the factory in separate components. This chapter describes the electric and pneumatic assembly of the components.

3.1 Safety instructions



Warning

Unexpected movements of the connected actuator technology and uncontrollable movements of loose tubing lines.

Risk of injury from electric shock, impact, crushing.

- Before working on the product, switch off the power supply and secure it against being switched on again.
- Before working on the product, switch off the compressed air supply, vent if necessary and secure it against being switched on again. Pneumatic system parts can still contain compressed air even when they are switched off.
  - Vent pneumatic system parts accordingly.
- When lifting the product, perform the following measures:
  - Secure movable components with transport locks to prevent unexpected movement.
  - Observe the product weight.
  - Adjust the lifting speed.
- When mounting the product, perform the following measures:
  - Secure the product in a professional manner.
  - Make sure the mounting surface is sufficiently strong to absorb the maximum forces.

3.1.1 Installation dimensions of the CPX terminal with controller and CPX node

The mounting holes on the end plates allow the CPX terminal to be secured to a flat surface.

The diagram below shows the mounting surface and the dimensions for the four threaded holes of screw size M6.

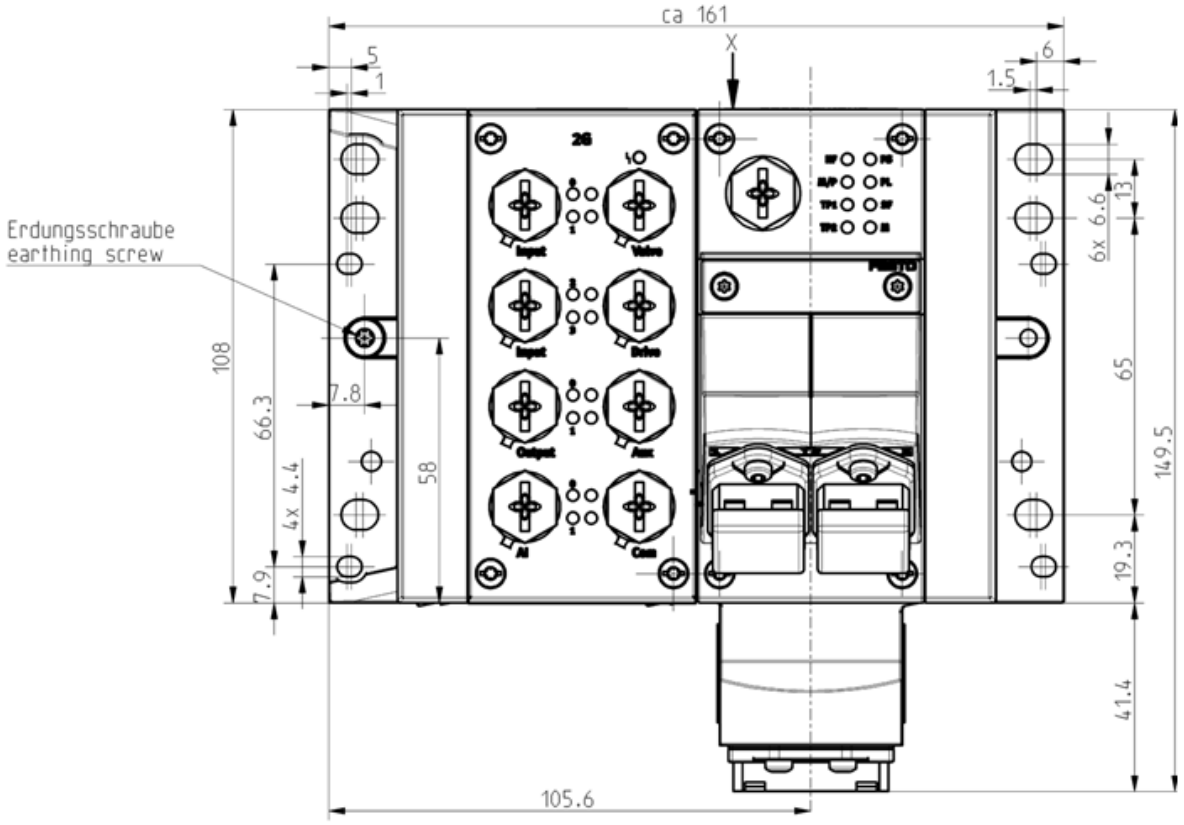


Figure 3-1 Installation dimensions of the CPX terminal with controller and CPX node

Further mounting options can be found in the CPX system description.

3.1.2 Installation dimensions of the control block

The control block can be mounted on a flat surface using 4 x M6 screws.

If vibrations or accelerations occur in the range of application, the control block must be mounted on rubber buffers.

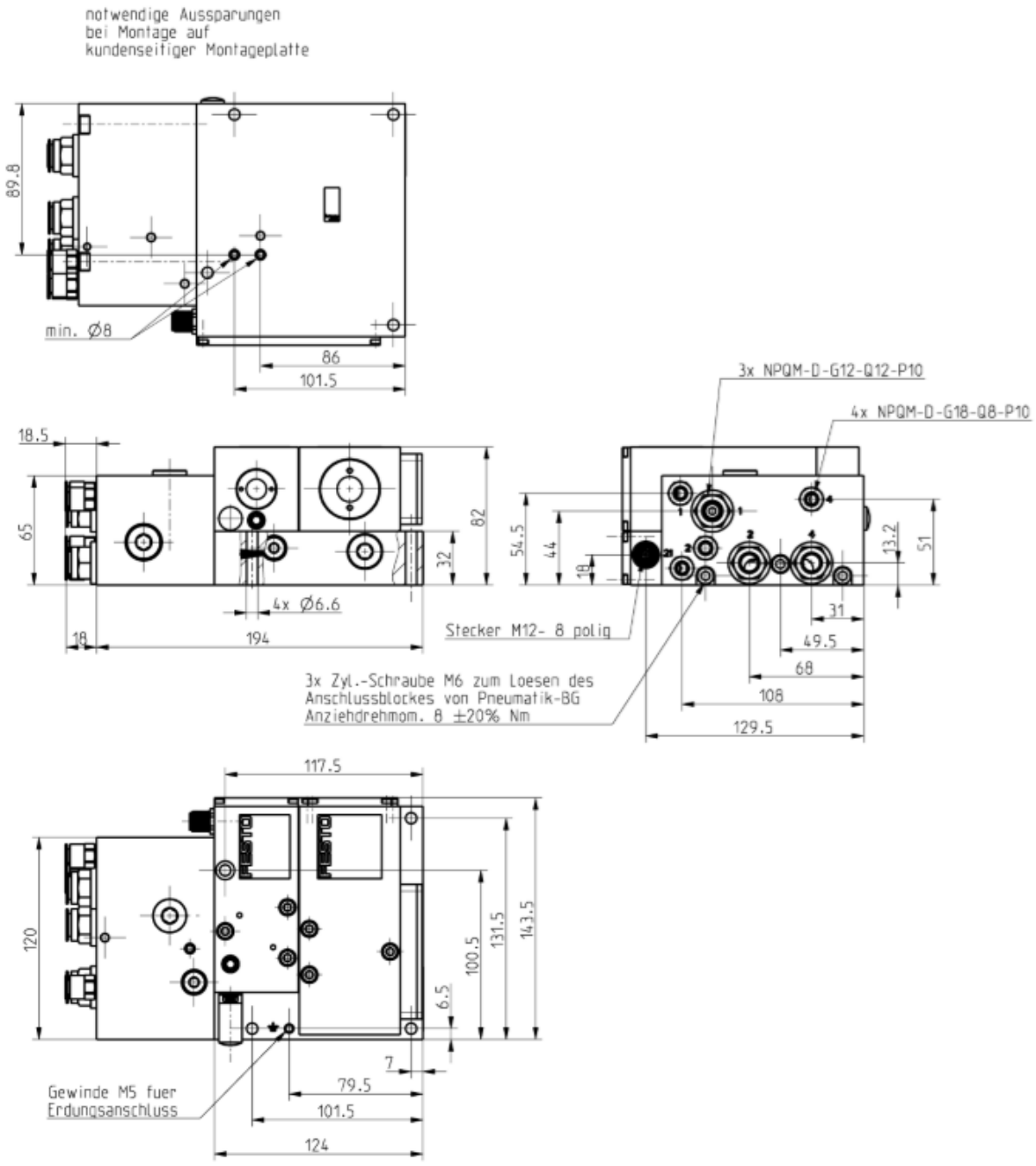


Figure 3-2 Installation dimensions of the control block with air supply shutdown

4. Installation

4.1 Pneumatic circuit diagram

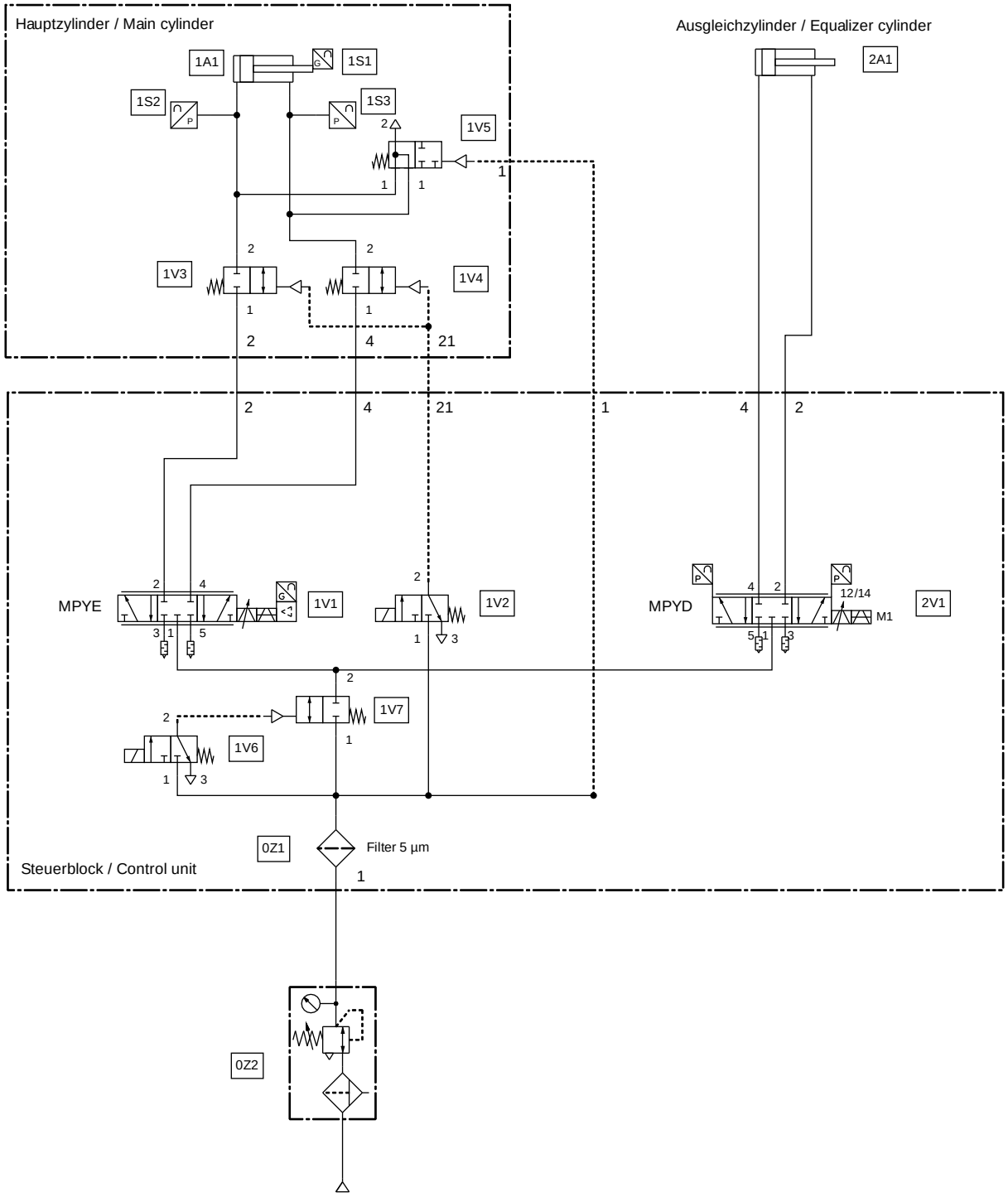


Figure 4-1 Pneumatic circuit diagram with air supply shutdown

4.2 Pneumatic installation

4.2.1 Main cylinder

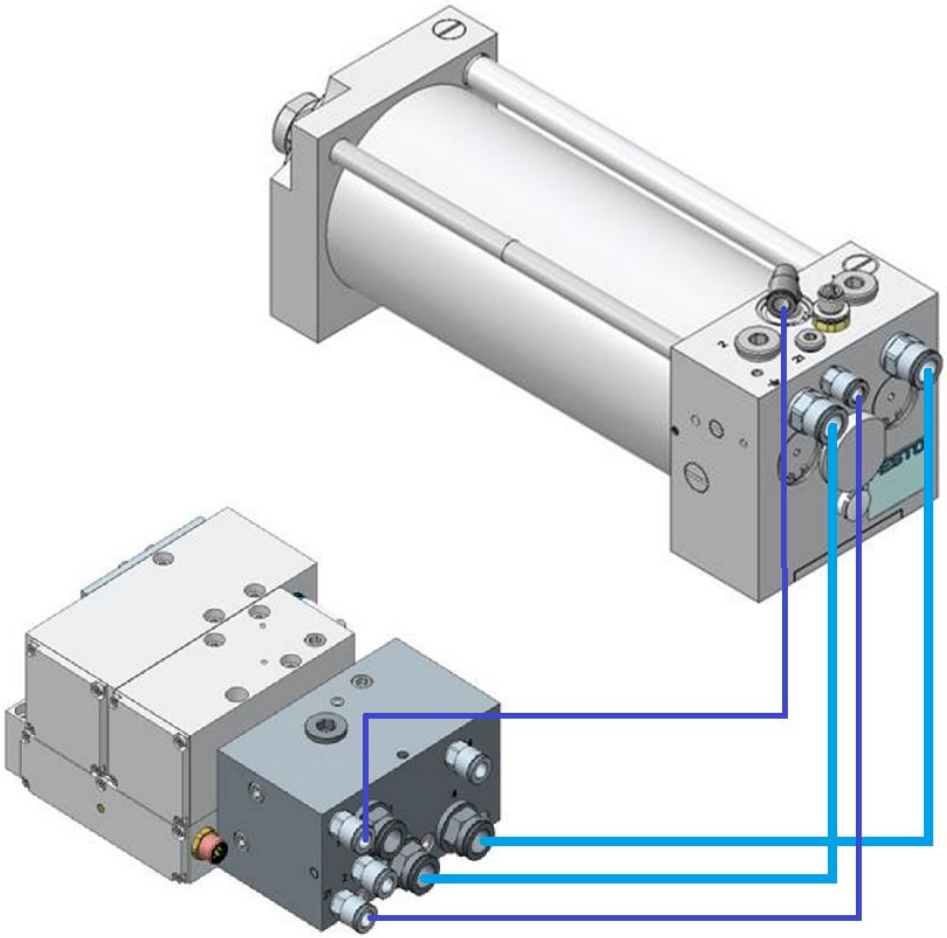
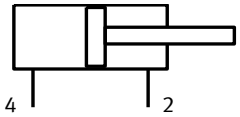
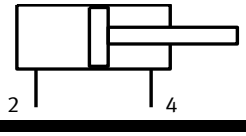


Figure 4-2 Tubing connection between the manifold block and main cylinder

4.2.2 Compensating cylinder

The tubing connection between the manifold block and the compensating cylinder varies depending on the design of the welding gun used.

Design	Pneumatic connection
C gun	
X gun	

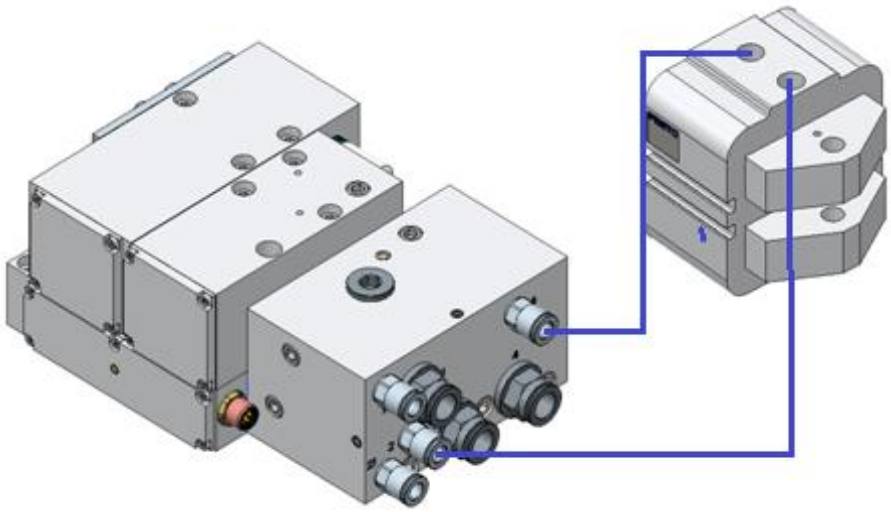


Figure 4-3 Tubing connection between the manifold block and compensating cylinder

- When connecting the compensating cylinder, please note the following:
  - the smaller differential pressure is set for movement to the welding piece
  - the greater differential pressure is set for movement to the stop

4.3 Electrical installation

4.3.1 Electrical connections on the controller

1 Input: 2 digital inputs [X1]
2 Input: 2 digital inputs [X2]
3 Output: 2 digital outputs [X3]
4 AI: Connection of external force sensor [X4]
5 Valve: Control block connection [X5]
6 Drive: Interface to the main cylinder [X6]
7 Service interface on the CPX bus node
8 System supply

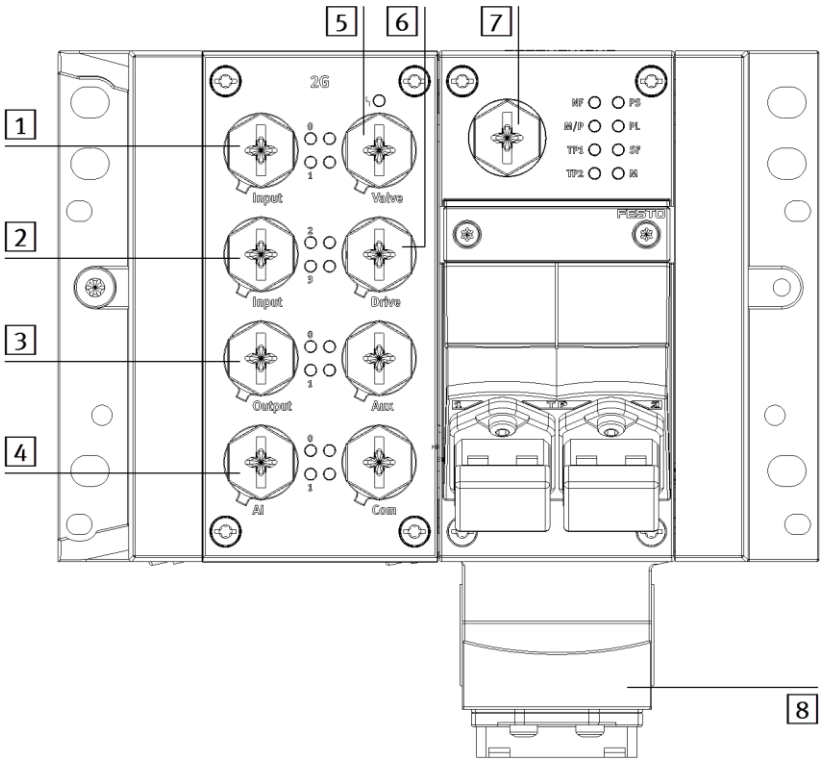


Figure 4-4 CPX terminal with CPX node and controller CPX-WELD-2G-SA

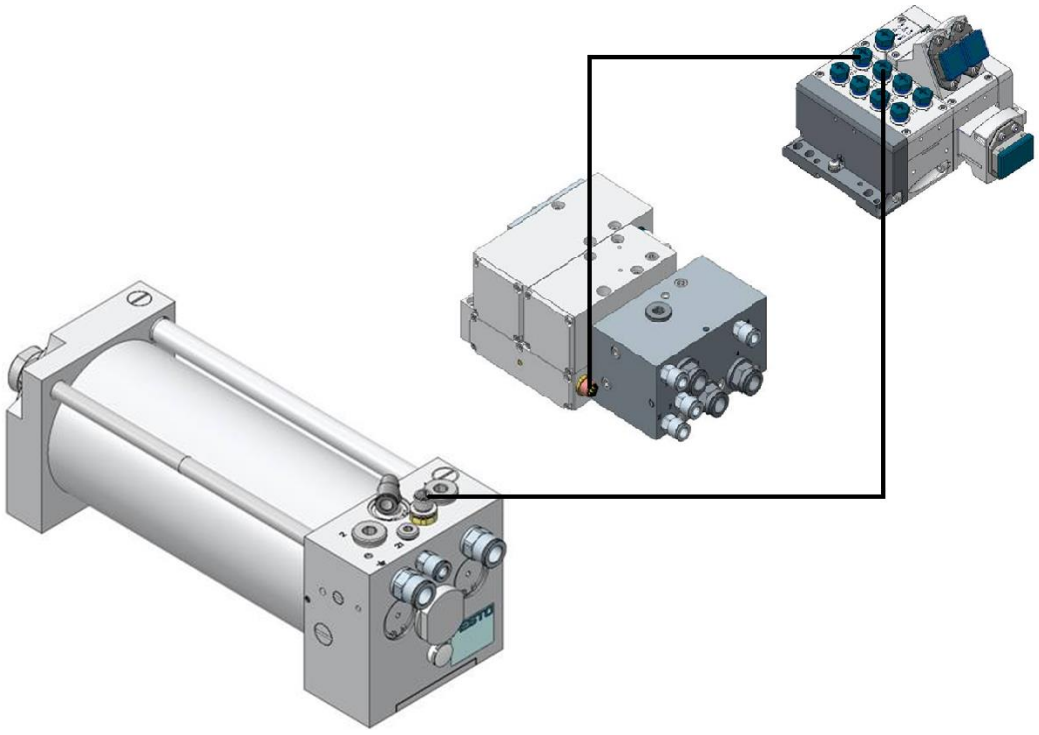


Figure 4-5 Electrical installation of welding gun system

Input: Digital inputs [X1] and [X2]

Input 0/Input 1 1		**Input 2/Input 3** 2	
Pin	Signal	Pin	Signal
1	24 V sensor supply	1	24 V sensor supply
2	Input x+1 (available input)	2	Input x+3 (available input)
3	0 V	3	0 V
4	Input x (available input)	4	Input x+2 (available input)
5	PE	5	PE

Output: Digital outputs [X3]

Output 0/Output 1 3	
Pin	Signal
1	n.c.
2	Output x+1
3	0 V
4	Output x (reserved)
5	PE

Valve: Valve block connection [X5]

Valve 5	
Pin	Signal
1	24 V valve supply MPYE
2	24 V valve supply MPYD
3	Control of air supply shutdown
4	0 V
5	CAN_H
6	CAN_L
7	Block control of cylinder 24 V load voltage supply US2
8	0 V load voltage supply US2

Drive: Connection of main cylinder [X6]

Drive 6	
Pin	Signal
1	24 V power supply
2	0 V
3	CAN_H
4	CAN_L
5	PE

Response when switching off the load voltage supply

The logic and load voltages are separated in the controller.

Following Power On, the internal closed-loop controller must be activated with a “1” on Enable. If Enable is then set to “0”, the position controller remains active, regulating to the current position. However it is not possible to initiate any travel jobs.

Switching off the load voltage supply US2 causes the closed-loop controller to be switched off. In such cases the additional pneumatic circuit is activated. The pressures in the drive are locked. When US2 is switched back on, the closed-loop controller will remain deactivated until Enable is set again and the additional pneumatic circuit is blocked.

AI: Connection of external force sensor [X4]

AI 4	
Pin	Signal
1	24 V sensor supply
2	Analogue input 1 (0 ... 10 V)
3	0 V
4	Analogue input 0 (0 ... 10 V)
5	PE

Interlinking block

Interlinking block with system supply 7	
Pin	Signal
1	24 VEL/SEN operating voltage supply for electronics/sensors
2	0 VEL/SEN operating voltage supply for electronics/sensors
3	24 VVAL/OUT load voltage supply for valves/outputs
4	0 VVAL/OUT load voltage supply for valves/outputs
5	FE

Basic information regarding the power supply concept of the CPX terminal can be found in the CPX system description.

5. Commissioning

5.1 Notes on commissioning



Warning
Risk of injury due to electric shock
- Use only voltage sources that ensure a reliable electric separation from the mains network in accordance with IEC 60204-1/EN 60204-1.
- For the electrical power supply, use only PELV circuits in accordance with IEC 60204-1/EN 60204-1 (Protective Extra-Low Voltage, PELV).

Before switching on the compressed air, you must set the parameters for the CPX-WELD-2G-SA.
The necessary settings can be implemented by using the software package WinSPAS.

The supply voltages are fed in at the fieldbus node.
- Use only the original cable.
- Do not place the cables near or parallel to cables with high noise.
- Seal unused connections with the protective caps supplied.
- Earth the system properly.

5.2 Procedure for commissioning

When commissioning for the first time, you will need the software package WinSPAS CPX, version 3.x.x.x or later. WinSPAS is not required after this initial setting of parameters.

5.2.1 Brief instructions for commissioning

1. Start the software package WinSPAS CPX.
2. Activate the online mode.
3. Start the commissioning assistant via the [Calibration] menu and follow the instructions.
4. Enter the number of the gun.
5. Select the welding gun from the database.
6. Perform a travel calibration.
7. Perform a force calibration.
8. Adapt the parameters under “System” (optional).

The welding gun is now ready to operate.

6. Operation via WinSPAS

6.1 System requirements

For operation of the WinSPAS software, you require a Pentium-compatible PC with the following minimum system requirements.

- Operating system Windows XP, 7, 10
- Recommended RAM: at least 1 GB
- 10 MB free space on hard disk
- CD-ROM drive
- A free USB slot and a USB adapter ➔ 6.7.1
  or alternatively
  a free Ethernet port for connecting the controller (for online mode only)

6.2 Installation

To install the software package WinSPAS on your hard disk, go through the following steps.

1. Start Microsoft Windows.
2. Place the installation CD in the CD-ROM drive.
3. Go to the Start menu and select the command [Run].
4. In the “Run” dialog box, enter the drive letter and the name of the installation program (e.g. D:\SETUP.EXE).
5. Start the installation program by pressing “OK”.
6. Follow the instructions in the installation program.

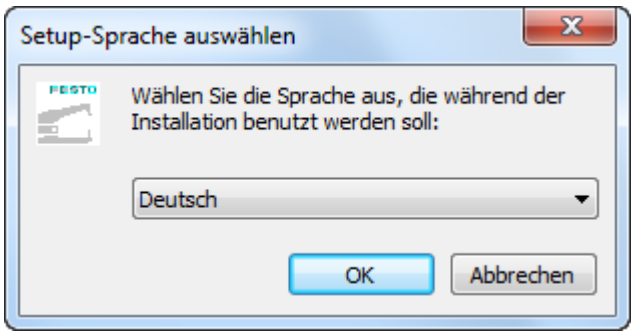


Figure 6-1 Language selection when starting the installation process

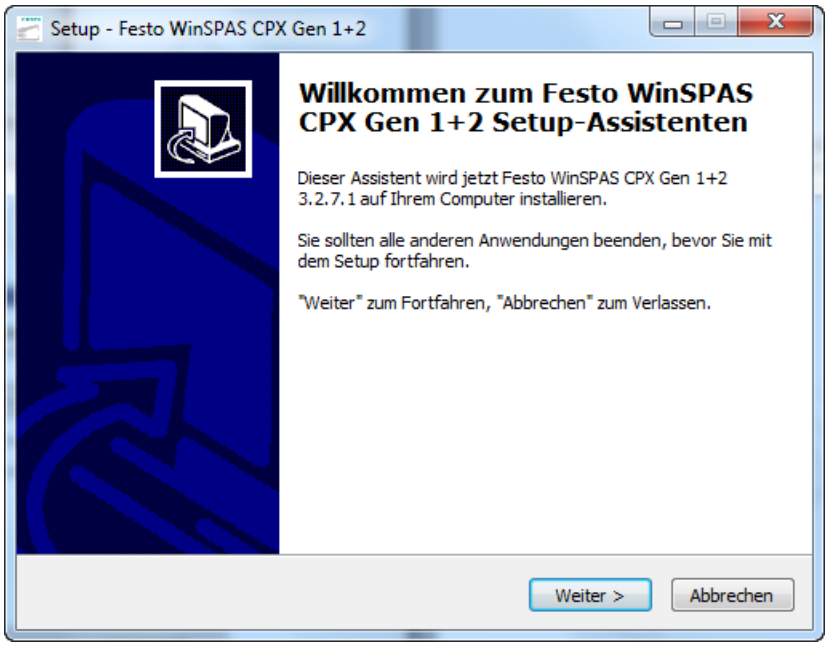


Figure 6-2 Startup dialog window for the installation process



Figure 6-3 Installation complete

7. Select “Start Festo WinSPAS CPX” and click [Finish].

6.3 Uninstallation

WinSPAS can be uninstalled by using the “Add or Remove Programs” function in the Windows Control Panel. Proceed here as described in your Windows manual.

6.4 Function range

WinSPAS is used for setting up welding gun systems.
The following functions are available.

- Project planning
- Commissioning
- Controller
- Diagnostics
- Simulation

6.5 Product range overview

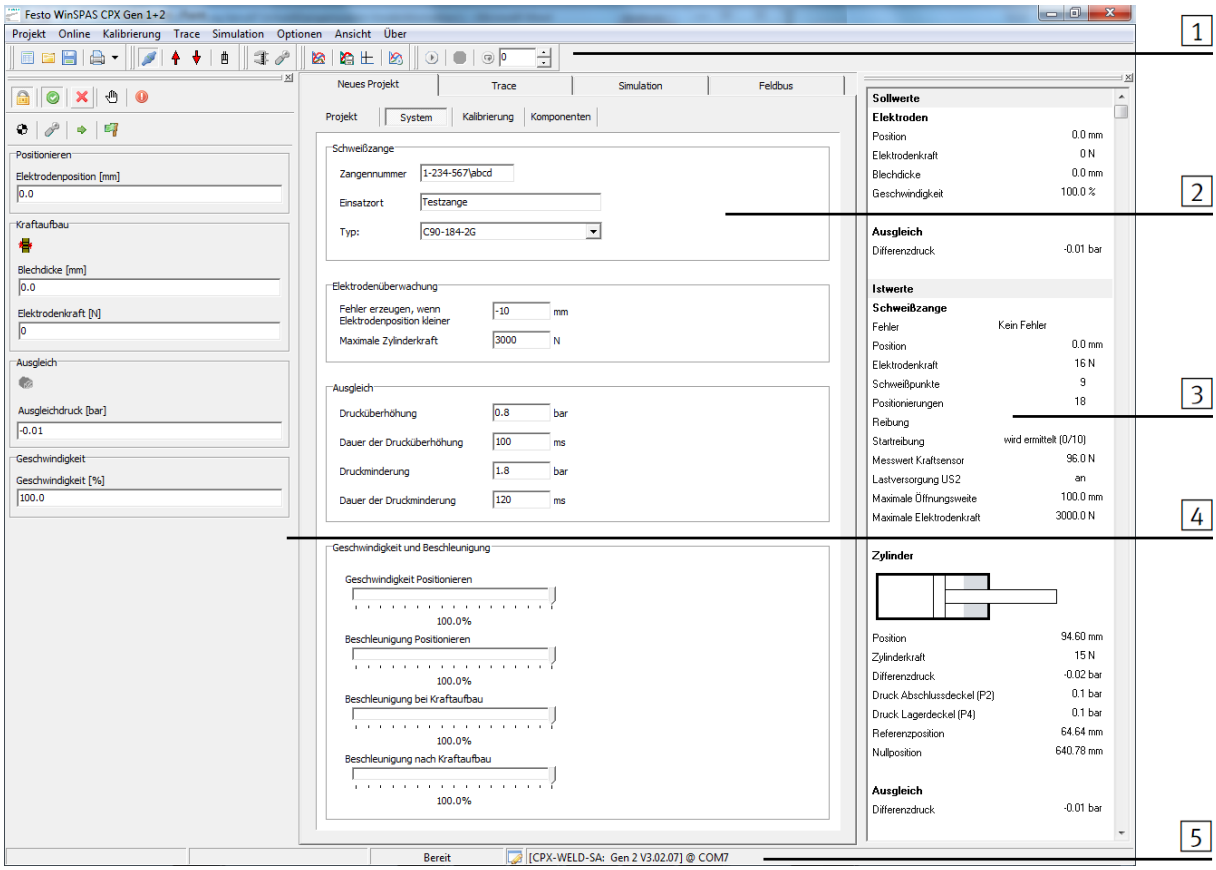


Figure 6-4 WinSPAS program window

Legend	Area/panel	Contents
1	Main menu and toolbar	In the main menu, you will find all the commands for general control of the program. The toolbar contains buttons for the most frequently used commands.
2	“Project”	This panel contains all of the information regarding the currently loaded project, along with the display of all components and parameters for the corresponding welding gun system.

Legend	Area/panel	Contents
3	“Setpoint/actual values”	This panel contains not only the setpoint values but also the actual values, as long as the connection to the controller of the welding gun drive is active.
4	“Controller”	This panel can be used to control the welding gun drive directly.
5	Firmware version	Connection display and display of the firmware version for the connected controller.

6.5.1 Menu commands

[Project] menu

The [Project] menu contains the commands for generating, saving and reloading projects, which contain all the relevant configuration, parameterisation and control data.

Command	Button	Function
[New]		Creates a new project file. The current project file will be closed after a prompt to save.
[Open]		Opens a saved project file; if there is an existing connection: permits system data synchronisation and transfer of the file contents to the controller of the welding gun drive.
[Save]		Saves the current project file. When the project file is saved for the first time, the “Save file as” dialog box will appear so that you can enter a name.
[Print]		Prints the tab open under System.

[Online] menu

The commands in the [Online] menu are used for establishing the online connection and for uploading and downloading projects.

Command	Button	Function
[Connect]	 	Establishes the connection to control the welding gun system. Clicking this menu command again terminates the active connection.
[Upload]		Starts the process of receiving project data from the controller. The data is transmitted from the controller to the PC.
[Download]		Starts the process of transmitting project data to the controller. The data is transmitted from the PC to the controller.

[Calibration] menu

The commands in the [Calibration] menu are used for calibrating the welding gun system.

Command	Button	Function
[Travel calibration]		Adjustment of the travel of the welding gun.
[Force calibration]		Adjustment of the force of the welding gun.
[Commissioning assistant]		Opens the commissioning assistant.

[Trace] menu

Command	Button	Function
[Export trace]		Exports the current trace to a file.
[Show crosshair]		Shows or hides the crosshairs.
[Show legend]		Shows or hides the legend.
[Reset zoom]		Sets the zoom factor to 100 %.
[Start trace recording]		Starts the trace recording immediately.

[Simulation] menu

Command	Button	Function
[Start]		Starts the simulation.
[Pause]		Interrupts the simulation.
[Repeat endless]		Repeats the simulation continuously, in an endless loop.

[Options] menu

Command	Function
[Welding guns]	Opens the welding gun database.
[Drives]	Opens the drive database.
[Firmware]	Opens the submenu for downloading firmware updates.
[Language]	Setting the programming language.

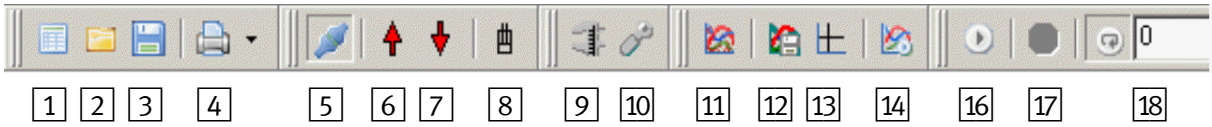
[View] menu

Command	Function
[Controller]	Opens / closes the “Controller” panel.
[Setpoint/actual values]	Opens/closes the “Setpoint/actual values” panel.
[Cylinder]	Opens the “Cylinder” dialog box.
[Last setpoint values]	Opens the “Last setpoint values” dialog box.

[About] menu

Command	Function
[Info]	Information about WinSPAS.

6.5.2 Toolbar of the program window



1 Create new project file for a welding gun system

2 Open project file

3 Save project file

4 Print

5 Establish/terminate the connection to the welding gun system – Online mode

6 Data transmission from the controller to the PC

7 Data transmission from the PC to the controller

8 Open the “Cylinder” dialog box

9 Travel calibration

10 Force calibration

11 Start trace automatically upon changes in the Control panel

12 Export trace

13 Show/hide crosshairs automatically

14 Start trace immediately

15 Start simulation

16 Pause simulation

17 Repeat simulation continuously in an endless loop

18 Number of simulation repetitions

Figure 6-5 Toolbar for quick access to important and frequently required functions

6.6 Show and hide the panels

The panels “Controller” and “Setpoint/actual values” can be displayed and hidden as required via the [View] menu.

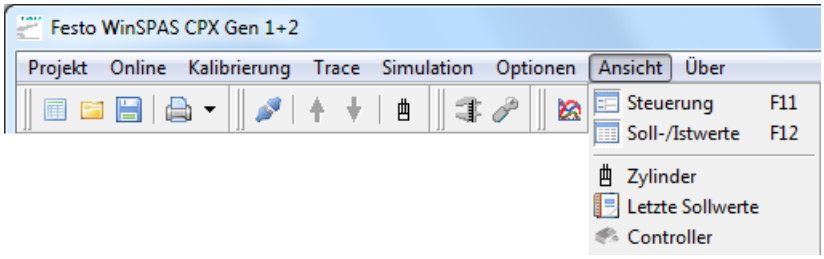


Figure 6-6 “View” menu commands

6.7 Online mode

For communication between WinSPAS and the controller CPX-WELD-2G-SA, you will need to activate the online mode. In this case, a connection is established between the PC and the controller via a virtual serial interface with the USB adapter NEFC-M12G5-0.3-U1G5 or, alternatively, via an Ethernet connection.

6.7.1 Connection between PC and CPX-WELD-2G-SA using a USB adapter

To connect the service interface on the CPX bus node to the PC, use the adapter NEFC-M12G5-0.3-U1G5 and a conventional USB cable with mini USB connector.

Information on installing the adapter and on the COMx interface used can be found in the documentation for the adapter.

1 Adapter
2 USB cable

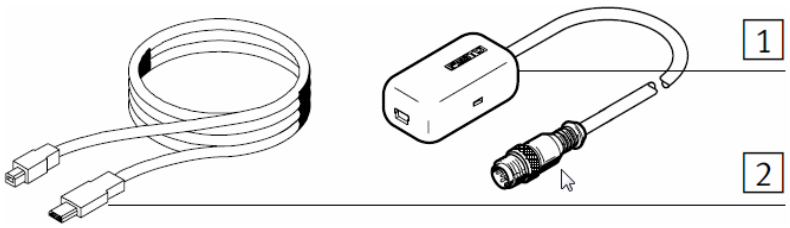


Figure 6-7 Adapter for connection to the PC

Note

The service interface (V.24) of the CPX terminal is not electrically isolated. It is not intended for permanent connection to PC systems or as a control interface.

- Remove the adapter during continuous operation.

Follow the instructions for operation and for installation of the required drivers on the accompanying CD-ROM or on the Internet:

➔ www.festo.com ➔ Downloads ➔ Download Area: Software, drivers and firmware ➔ Enter search term: FMT

- After commissioning, seal the service interface of the CPX node with the protective cap supplied.

6.7.2 Activating/deactivating the online mode

Once you have connected the PC and CPX–SPC–WELD, select [Connect] in the [Online] menu or click on “Establish connection” on the toolbar.

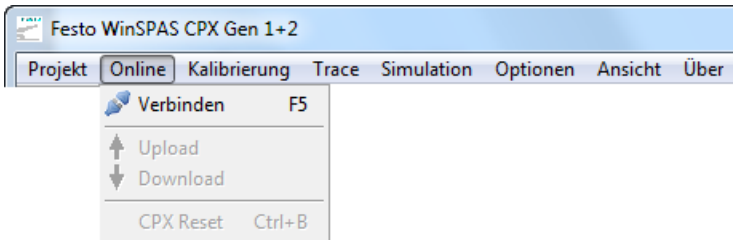


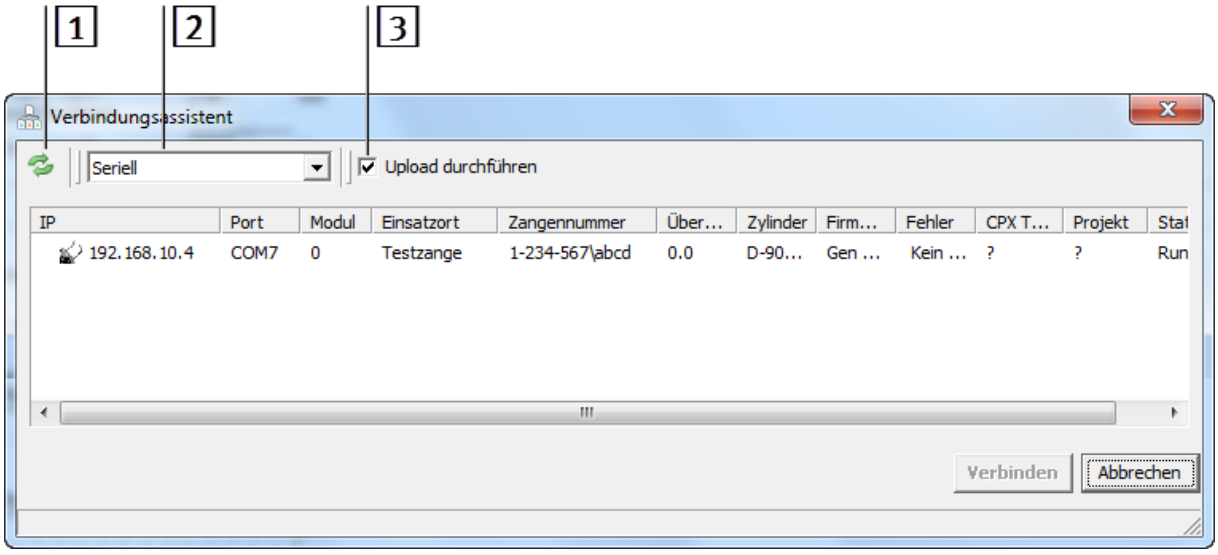
Figure 6-8 Online mode

The connection wizard is opened.

6.7.3 Connection wizard

The connection wizard searches for all of the controllers that are connected to the PC. Select the controller you wish to connect to and then click on [Connect].

If “Perform upload” is activated, the project data will be transmitted when the connection is set up.



1	Update	The network is re-imported
2	Protocol	Selection of the communication protocol (Profinet or Serial)
3	Perform upload	Triggers an upload when connecting

Figure 6-9 Connection wizard

The connection wizard shows the following data for the connected controllers:

– IP address (for Profinet controllers only)

- COM port (only if the “Serial” protocol has been selected)
- Module number
- Place of action
- Number of gun
- Gun ratio
- Cylinder
- Firmware version of the controller
- Current error message
- Firmware version of the node

6.7.4 Upload and download

[Upload] and [Download] are used for receiving and transmitting project data, respectively.

The following applies from the point of view of the PC.

Command	Button	Function
[Upload]		Receive project data from the controller.
[Download]		Transmit project data to the controller.

6.8 Project planning (components of a project)

A project consists of:

1 Application-specific data (calibration values, etc.)

2 Welding gun data (closed-loop controller parameters, drive used)

3 Drive data (diameter, length, ... of the main cylinder)

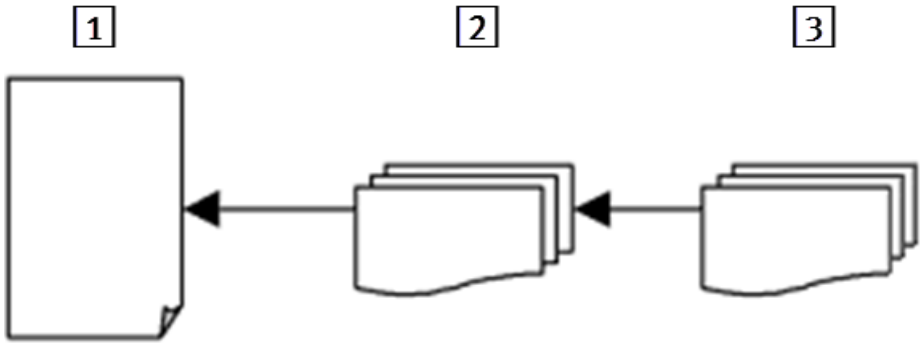


Figure 6-10 Components of a project

6.9 Project window

6.9.1 Project

The specifications in the project window are optional and are intended for the purpose of documenting the project. They are only saved in the *.spas file and not in the controller.

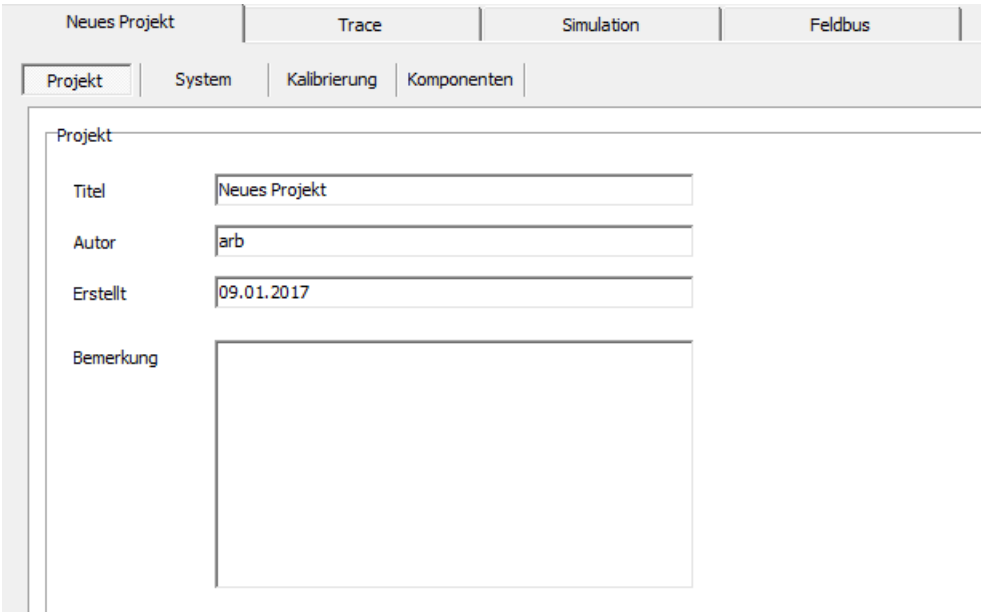


Figure 6-11 Project window – Project

Parameter	Significance
Title	Freely selectable title of the welding gun project.
Author	Your name.
Created by	Date of creation (set by default to the current date).
Comments	Any text can be entered here.

6.9.2 System

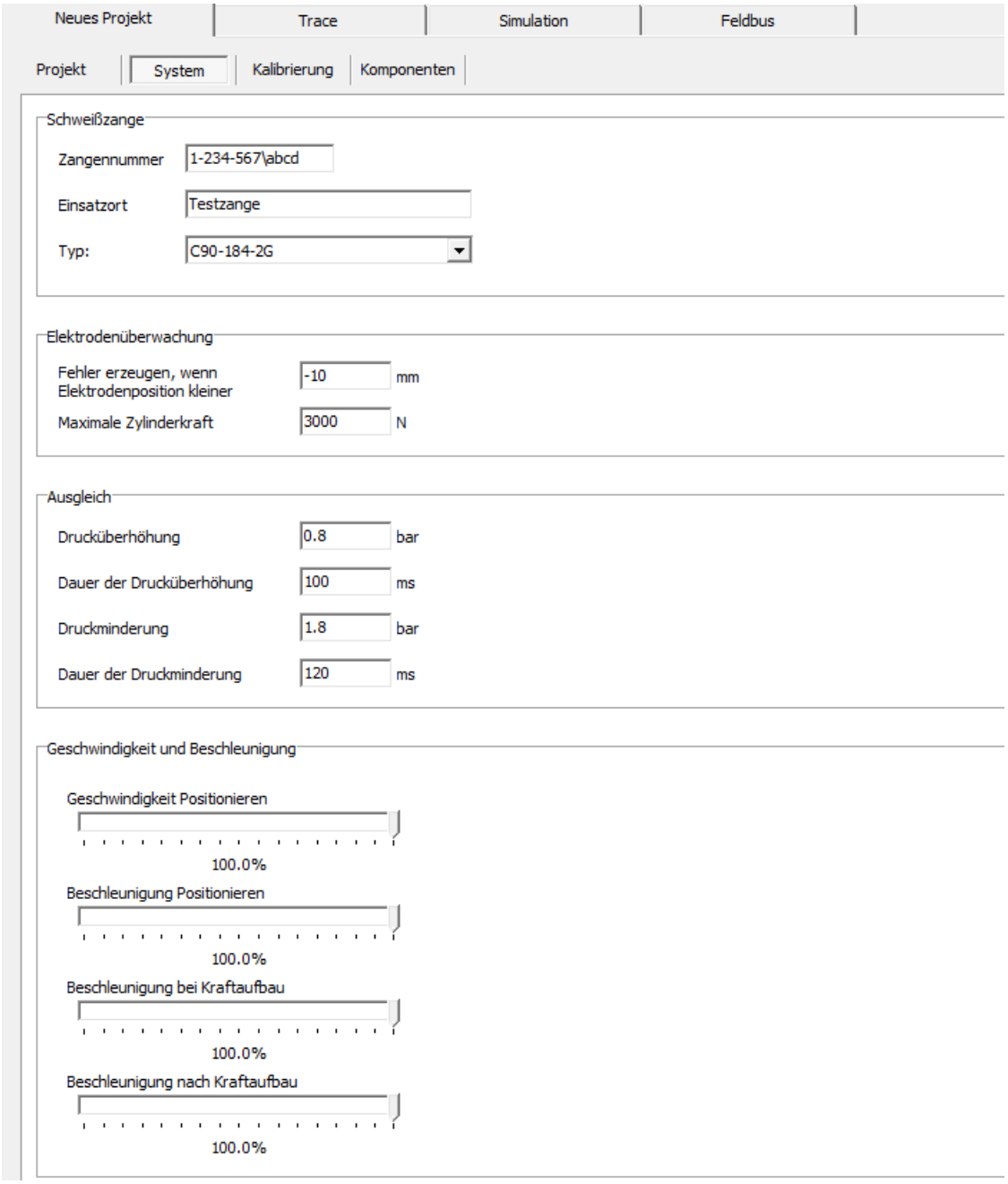


Figure 6-12 Project window – System

Welding gun

Welding gun parameter	Significance
Number of gun	Specification of the welding gun number.
Place of action	Specification of the location where the welding gun is used.
Type code	Selection of a parameter set from the welding gun database. This selection determines the cylinder used and the closed-loop controller parameters.

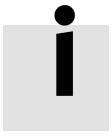
Electrode monitoring

Command	Button	Function
Report error when electrode position is less	mm	If an electrode position (relative to the reference position) less than this value is reached during force build-up, an error is generated. Protects against electrode breaks/cap loss.
Maximum cylinder force	N	Specifies the maximum cylinder force. An error is generated if exceeded during operation and if a greater force is required.

Compensation

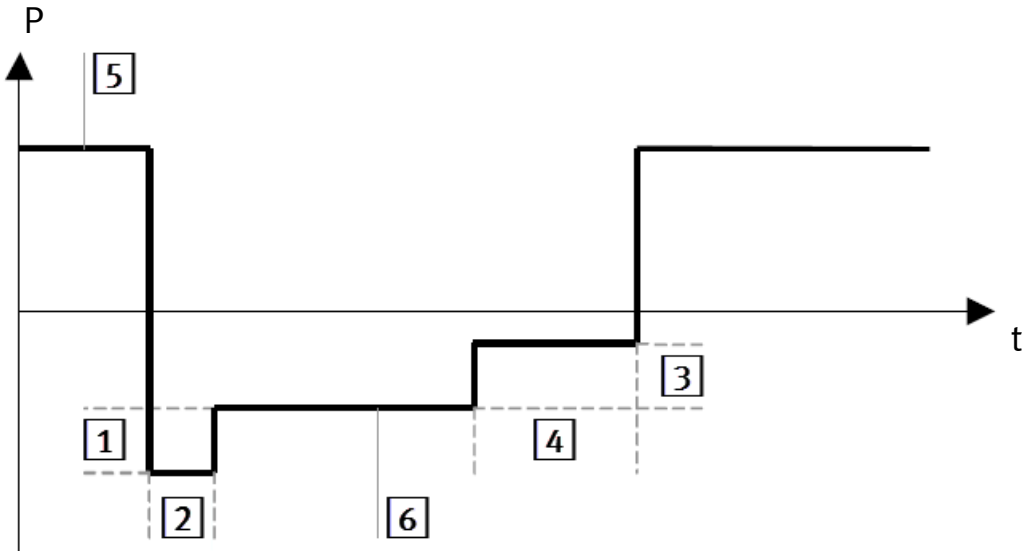
Two functions have been implemented for control of the compensating cylinder.
- Brief differential pressure overshoot for smooth break-away and movement to the welding piece
- Brief pressure reduction for movement to the stop (“soft landing”)

- Therefore, connect the compensating cylinder in such a way that the smaller compensating pressure must be set for movement to the welding piece, and the greater compensating pressure must be set for movement to the stop.



The overshoot or reduction is generated in the case of pressure changes of 1 bar or greater. Teaching of the compensation is therefore not affected.

Parameter	Unit	Significance
Rise of pressure	bar	When the compensating pressure is changed in a negative direction ($P_{setpoint} < P_{actual}$): Activated pressure: $P_{compensation} = P_{setpoint} - P_{rise}$
Duration of rise of pressure	ms	Time for which the pressure overshoot is in effect.
Pressure reduction	bar	When the compensating pressure is changed in a positive direction ($P_{setpoint} > P_{actual}$): Set pressure: $P_{compensation} = P_{setpoint} + P_{reduction}$
Duration of pressure reduction	ms	Time for which the pressure reduction is in effect.



1 Pressure overshoot
2 Duration of overshoot
3 Pressure reduction
4 Duration of pressure reduction
5 Compensating pressure at the stop
6 Compensating pressure at the welding piece

Figure 6-13 Pressure overshoot and pressure reduction

Speed and acceleration

Parameters for the speed and accelerations of the main cylinder can be set, in order to reduce momentum when moving large loads, for example.

Parameter	Significance
Speed (global)	Speed of all drive movements.
Acceleration (global)	Acceleration during all movements of the drive.
Acceleration at force build-up	Acceleration when force build-up is activated (movement towards the welding piece).
Acceleration after force build-up	Acceleration during positioning which follows force build-up (opening of gun arms after welding).

Example A

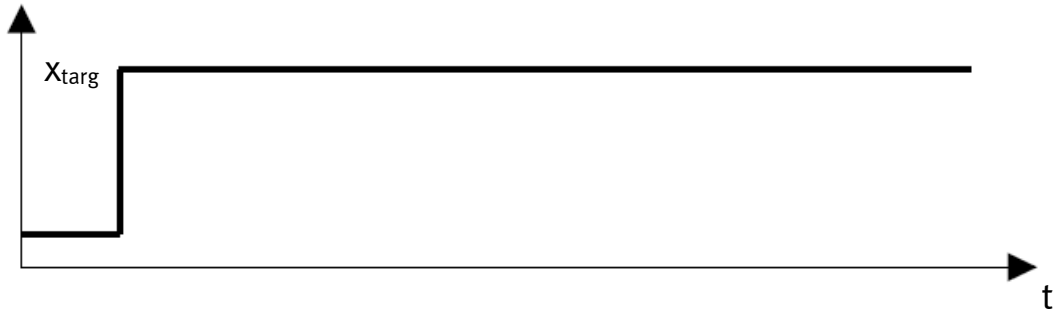


Figure 6-14 Maximum speed and acceleration (v = 100 %; a = 100 %)

Example B

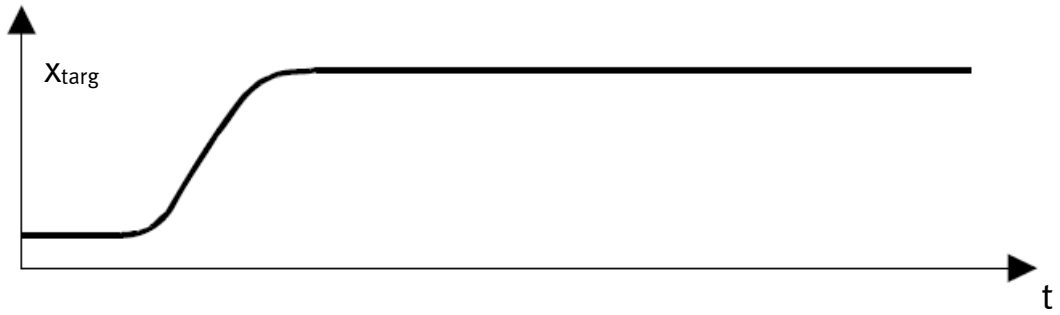


Figure 6-15 Reduced speed and acceleration (v < 100 %; a < 100 %)

6.9.3 Calibration

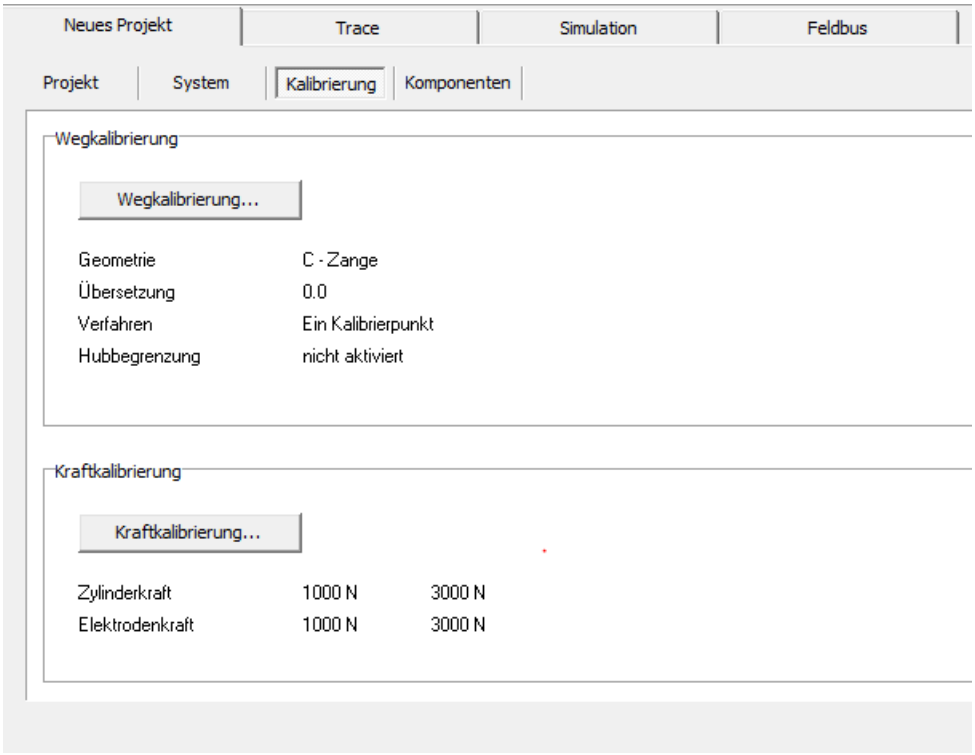


Figure 6-16 Calibration

Travel calibration

Command	Unit	Significance
Travel calibration button	–	Opens the “Travel calibration” dialog box.
Geometry	–	Gun geometry (X or C gun).
Ratio	–	The transmission ratio determined.

Force calibration

Command	Unit	Significance
Force calibration button	–	Opens the “Force calibration” dialog box.
Cylinder force	N	The cylinder force specified in the force calibration.
Electrode force	N	Measured electrode force.

6.9.4 Component diagnostics

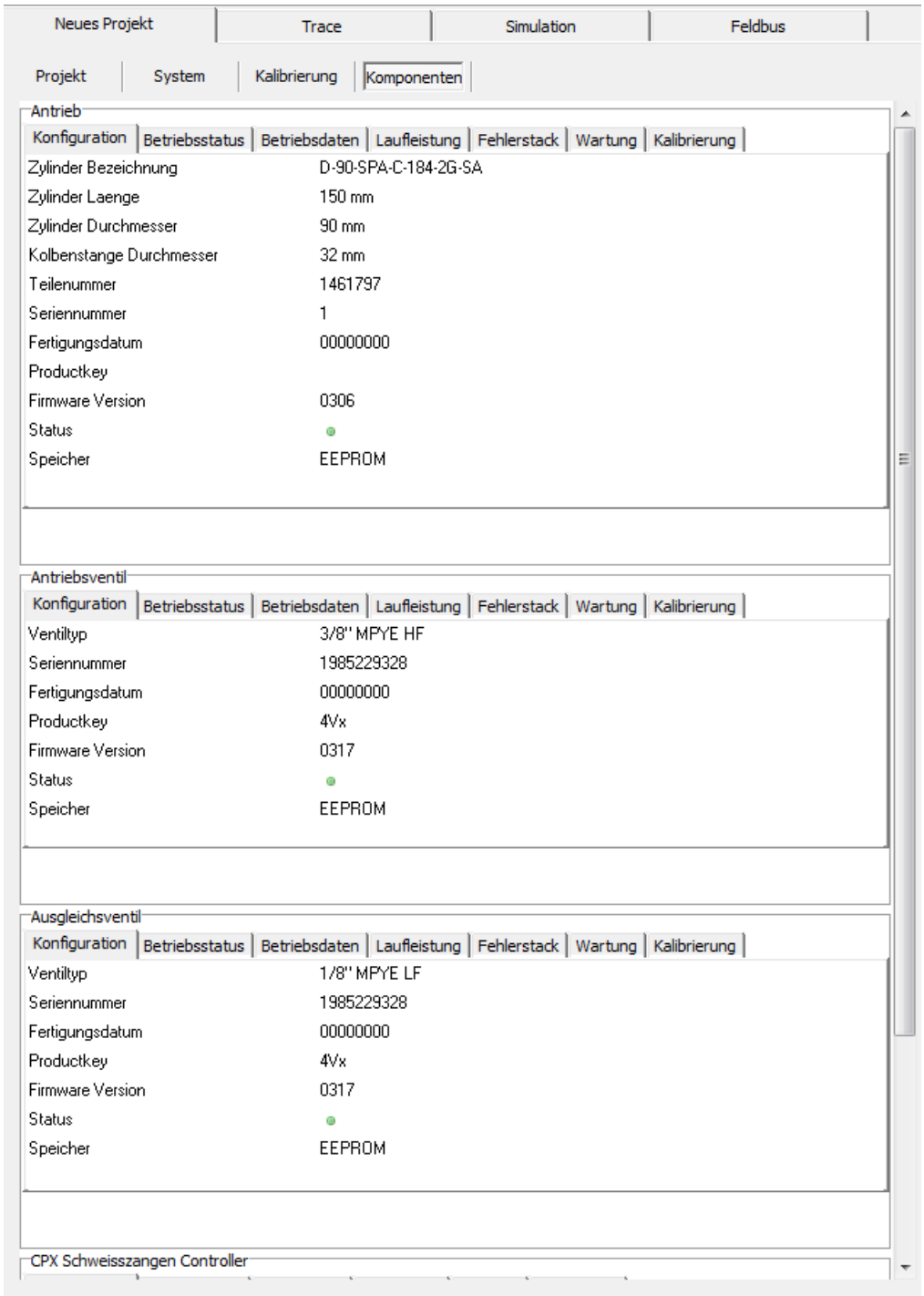


Figure 6-17 Component diagnostics

The system features extensive component diagnostics. Component data, service life and operating data remain stored in the components even if a power failure occurs; this information, in addition to the operating status for all system components, can be read and displayed in this dialog window.

6.9.5 Parameters in the Expert dialog box

The Expert dialog box is hidden when the program starts.

To activate the expert mode, position the mouse pointer over the [Connect] button in the toolbar ➔ 6.5.2, Figure 6-5, 5. Hold down the CTRL key and right-click your mouse.

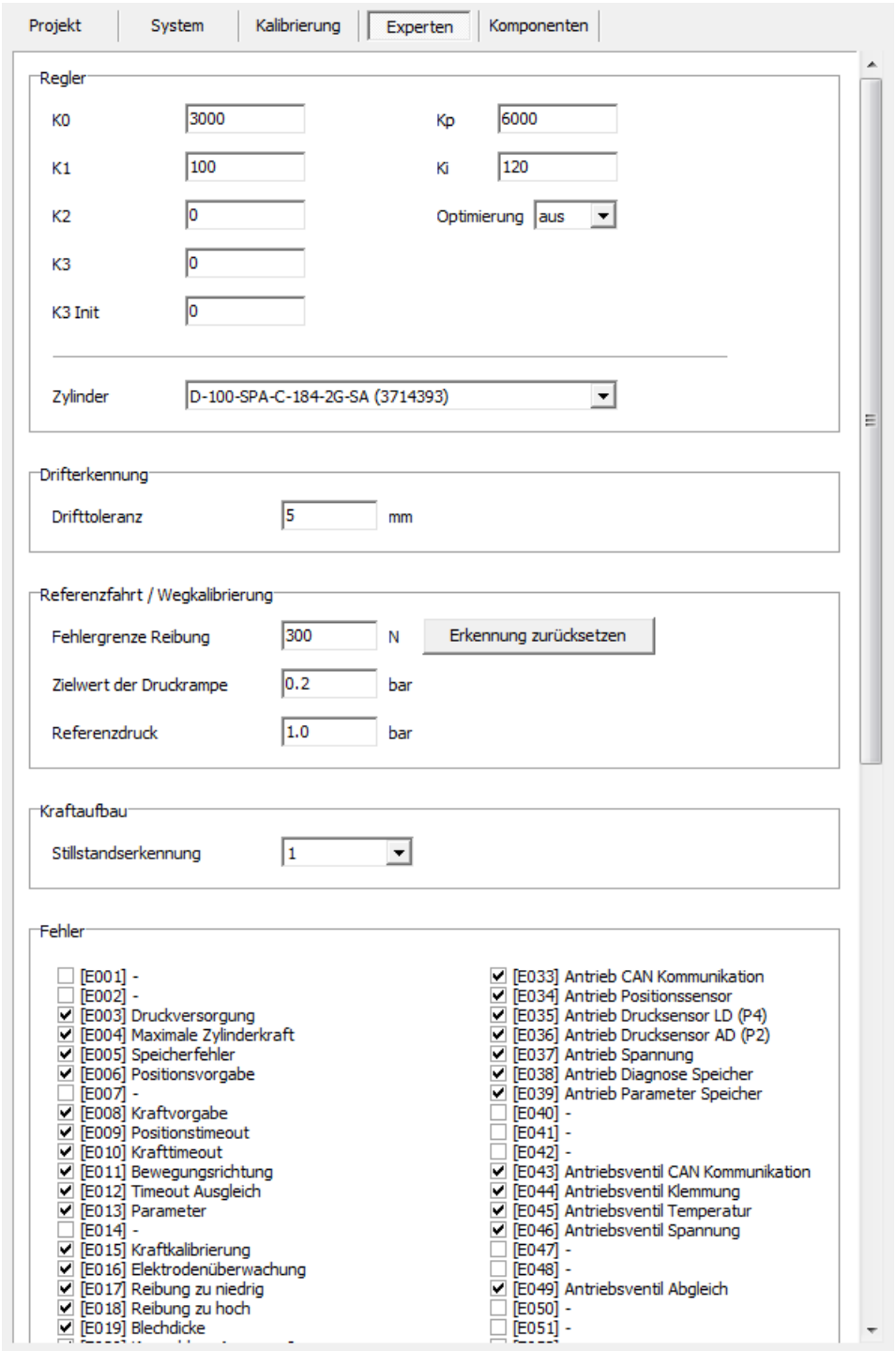


Figure 6-18 Expert dialog box

Controller

Parameter	Significance
K0, K1, K2, K3 and K3 init	Gain factors of the closed-loop position controller.
Kp and Ki	Amplification gains of the force and pressure regulator.
Valve offset	Offset for valve.
Cylinder	Drive used (from the database). Contains length and diameter information.

Drift tolerance

Parameter	Significance
Drift tolerance	Specifies the maximum permitted drift in mm while the additional pneumatic circuit is active. The drive can drift due to leakage. If the drift exceeds the specified limit, the “Setpoint/actual position comparison” bit on the fieldbus is reset. In such cases, the higher-order controller must ensure when restarting that the drive is in the specified setpoint position before any robot movement is carried out.

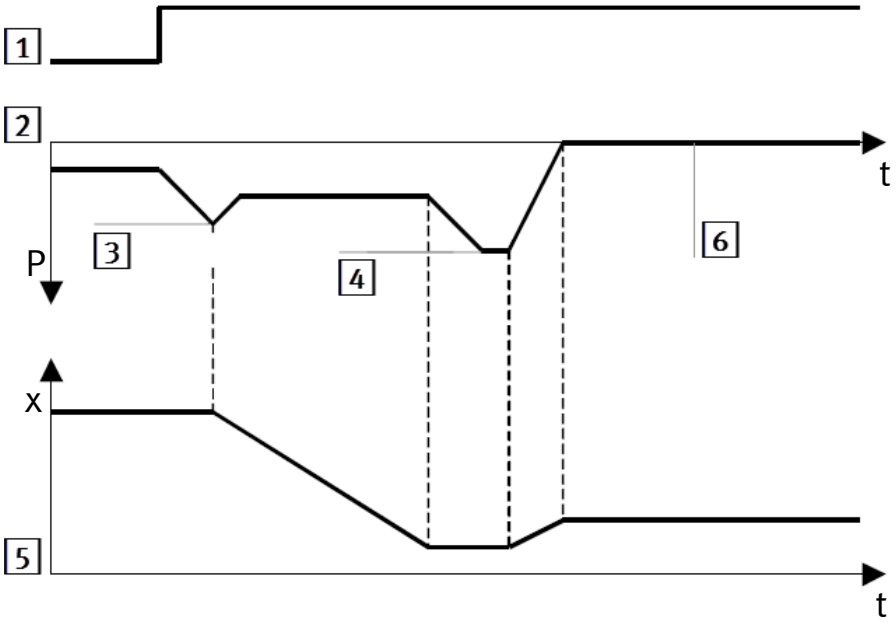
Homing

Command	Button	Function
Friction limit	N	If the friction measured during homing exceeds or falls below this value, an error is generated.
Target pressure	bar	Final value of the pressure ramp.
Reference pressure	bar	Trigger pressure identifying the point at which the gun arms are closed.

Friction detection

The controller checks the friction of the system during the homing procedure.
After processing the travel calibration, the break-away pressures measured during the next 10 homing runs are averaged. For every homing run after that, the current break-away pressure is measured and compared with this average.
If the current break-away pressure deviates from the set value, an error is generated.

The welding gun should always be in the same spatial position when homing and should always be started from the same position.



1 Homing
2 Pressure profile in the main cylinder
3 Break-away pressure
4 Reference pressure
5 Gun position
6 Target pressure

Figure 6-19 Friction detection

The system friction value can be reset with the “Reset detection” button.
The system friction is then determined again.

Force build-up

Standstill detection: During force build-up, the gun arms could move due to deflection of the gun. The standstill detection value determines the limit value below which the speed must fall in order for the “Force reached” signal to be generated.

The following applies:

Standstill detection	Severity level
0	Detection switched off. “Force reached” is output as soon as the welding force is reached.
1 ... 5	Detection active. The “Force reached” signal is output as soon as the welding force is reached and the speed of the drive falls below the limit value.

The permitted limit falls with increasing value. The “Force reached” signal is output later as the filter factor rises.

Error

The generation of particular errors can be prevented by deactivating the corresponding check boxes.

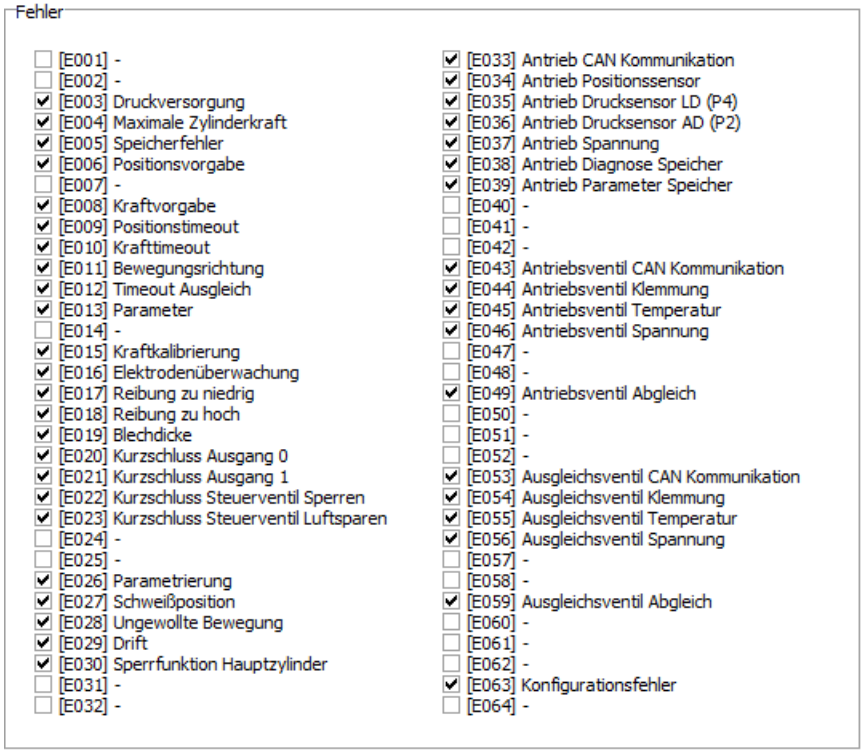


Figure 6-20 Expert dialog box – Errors

Warning
Risk of injury.
The welding gun can perform uncontrolled movements.
- Deactivate errors only when necessary for your application.

Diagnostic function for blocking valves

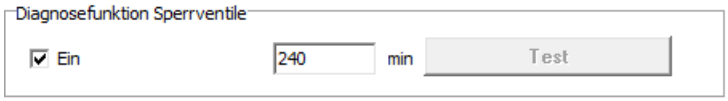


Figure 6-21 Diagnostic function

The diagnostic function (DF) is used to check the blocking function of the main cylinder. It enables a possible malfunction of the blocking valves to be detected in the main cylinder. Error E30 will be issued in this case, because the safe stop of the main cylinder is no longer guaranteed.

- Activate the diagnostic function in the expert dialog box.
- Set a period for the cyclic repetition of the diagnostic function.
- Use the “Test” button to run the DF at any time.

When activated, the DF is invoked after the preset time has elapsed. The DF will then be executed after completion of the next homing run.
As a result, the interface between the robot and controller remains unchanged.

If the DF is terminated with an error, error number E30 will be issued and the DF will be restarted at the next homing run.

Test function for data acquisition

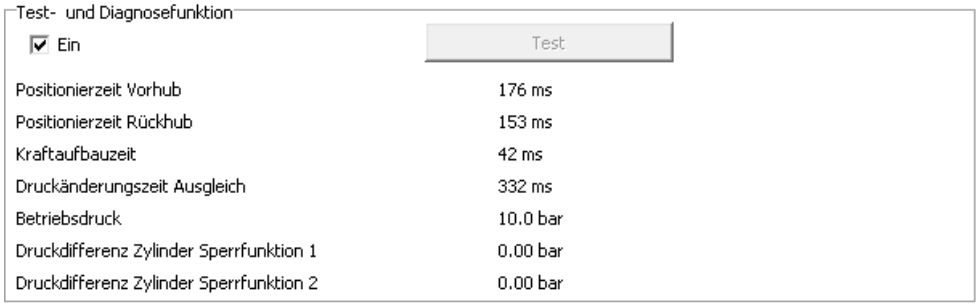


Figure 6-22 Test function for data acquisition

You can request a system test function by using a PC software tool. When activated, the test function will be executed after completion of the next homing run.

The test function checks all of the relevant gun functions and determines dynamic data.

- Use the “Test” button to run the test function at any time.

6.9.6 “Drives” database

This database stores all specifications for the main cylinders.

- Open the “Drives” dialog box via the menu [Options] [Drives].

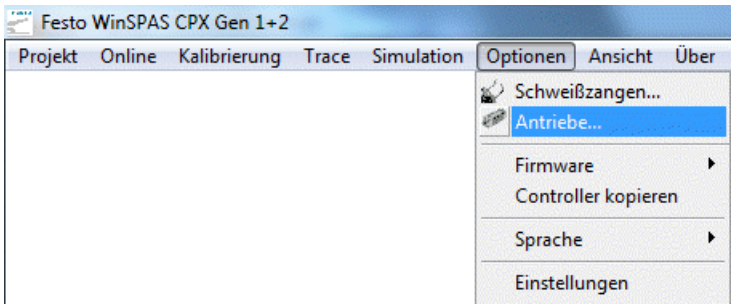
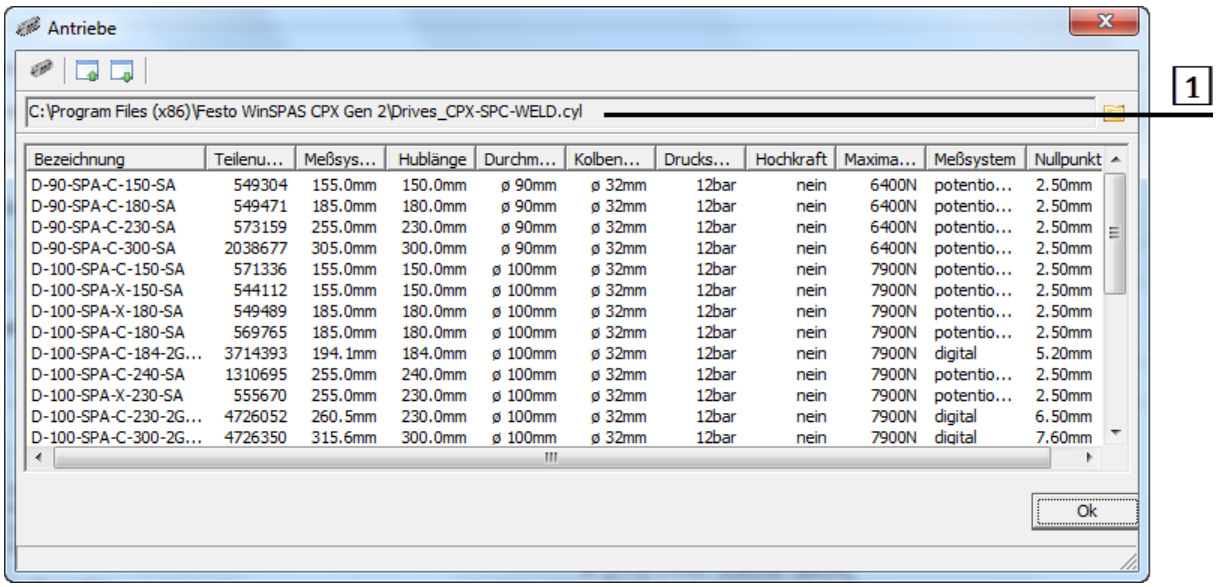


Figure 6-23 [Drives] menu command



1 Path where the drive database is saved

Figure 6-24 “Drives” database

- Double-click an existing drive to correct its key features

Command	Button	Function
[Create new]		Opens a dialog for entering the characteristics of a new drive.
[Import]		Imports a file with the key features of a drive.
[Export]		Exports the drive database.
[Open]		Opens a saved database for drives.
[OK]	–	Applies a drive from the database to the project.

The path where the database is saved can be chosen freely by the user. That allows different computers to access a shared database, for example.

Creating a new drive

- In the “Drives” dialog box, select the “Create new” button.

You will see a dialog box where you can enter all the properties of the new drive.

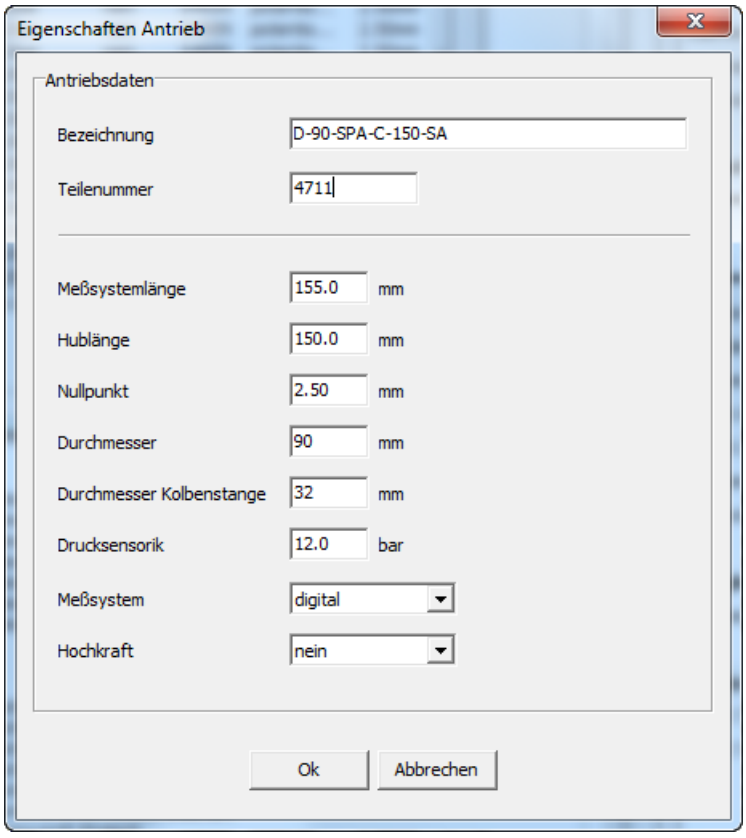


Figure 6-25 “Drive key features” dialog box

Input field	Contents
Designation	Name of the cylinder; any name can be chosen.
Part number	Part number of the cylinder ➔ Product label.
Measuring system length	Length of the built-in measuring system.
Stroke length	Length of the mechanical stroke.
Diameter	Diameter of the piston.
Diameter of the piston rod	Diameter of the piston rod.
Pressure sensors	Effective range of the built-in pressure sensors.
Measuring system	Type of measuring system.
High force	Indicates whether the cylinder is a high-force cylinder.

Importing a drive

The “Import drive” function can be used to import a cylinder that is saved in a different database into the current database.



1. In the “Drives” dialog box, click the “Import” button.
2. In the file selection dialog box, select a cylinder database.
3. In the “Import drive data” dialog box, select the drives to import and confirm their transfer to the drive database by pressing [OK].

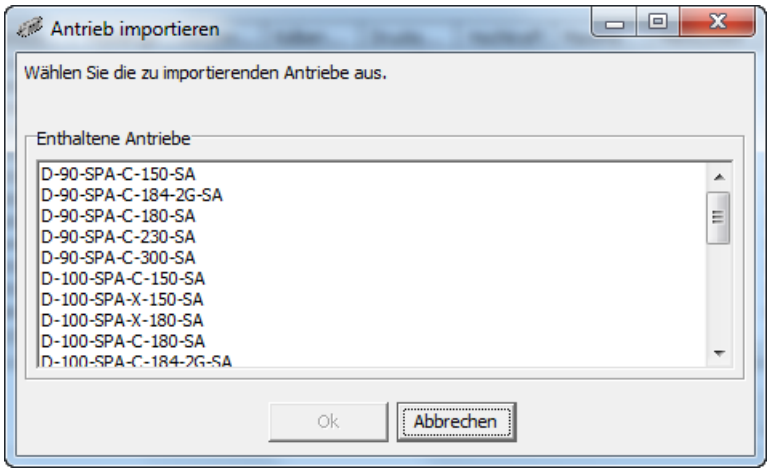


Figure 6-26 “Import drive data” dialog box

6.9.7 “Welding guns” database

This database stores all parameters for the welding guns.

A welding gun parameter set includes the following:

- Closed Loop
- Cylinder (with length, diameter, ...)
- Geometry

Parameters for identically constructed guns only need to be entered once.

1. Open the “Parameterisation of welding guns” dialog box by selecting the [Options] menu and clicking [Welding guns].

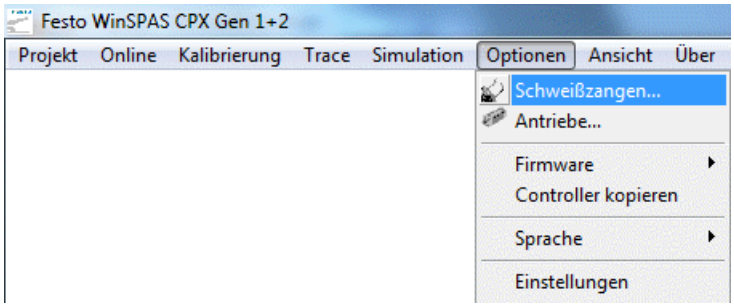
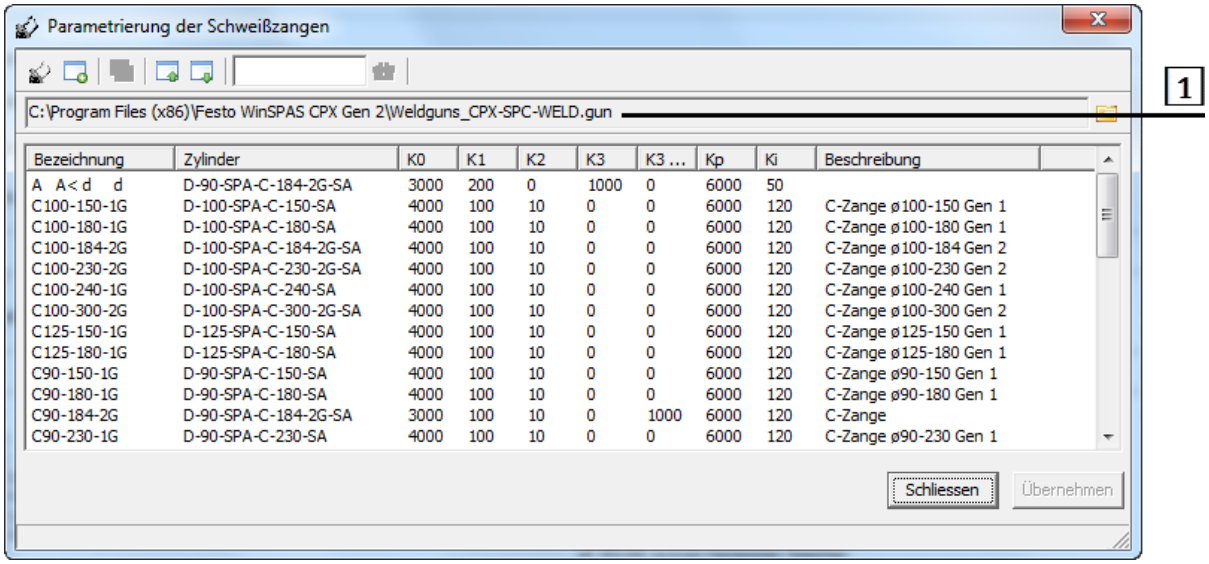


Figure 6-27 Menu command [Welding guns]

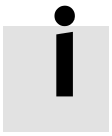


1 Path where the welding gun database is saved

Figure 6-28 “Parameterization of welding guns” database

2. Select a welding gun from the database and confirm your selection by pressing “Apply”.
3. Double-click a welding gun to edit the parameters.

Command	Button	Function
[Create new]		Opens a dialog for entering the parameters of a new welding gun.
[Import from project]		Creates a welding gun using the current project data.
[Import]		Imports a file with the parameters of a welding gun.
[Export]		Exports the welding gun database.
[Open]		Opens a saved database for welding guns.
[Apply]	–	Applies a welding gun from the database to the project.
[Close]	–	Closes the dialog box without applying the welding gun parameters to the project.



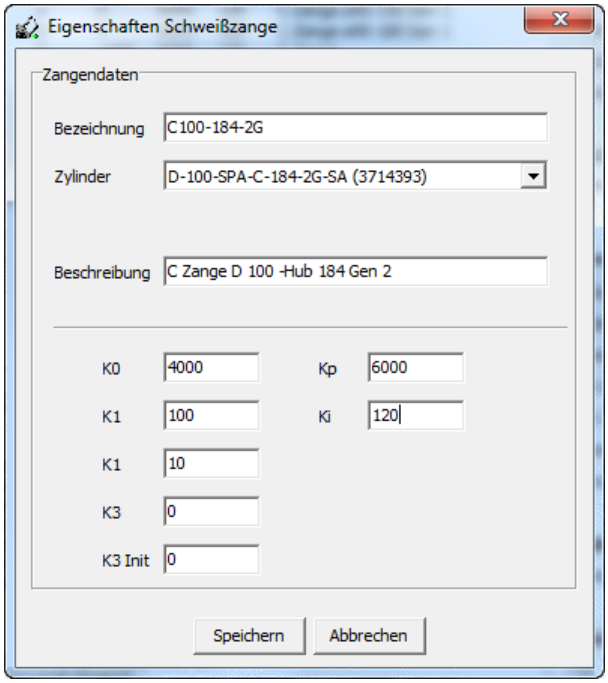
The path where the database is saved can be chosen freely by the user. That allows different computers to access a shared database, for example.

Creating a new welding gun



- In the “Parameterisation of welding guns” dialog box, click the “Create new welding gun” button.

The “Welding gun key features” dialog box appears, which allows you to enter all of the parameters for the new welding gun.



Input field	Contents
Designation	Freely selectable designation of the welding gun (max. 32 characters).
Cylinder	Select the drive built into the welding gun from the database.
Geometry	Select the type of welding gun geometry.
Description	Freely selectable description of the welding gun.
K0, K1, K2, K3, K3 Init	Control factors of the position controller.
Kp, Ki	Control factors of the force controller.

Figure 6-29 “Welding gun key features” dialog box

6.10 Travel calibration

The travel calibration is used to determine:
- the zero position
- the transmission ratio for X guns

Travel calibration is carried out independently of the gun geometry (C/X).

6.10.1 Zero position and reference position

All position specifications (both fieldbus and in WinSPAS) refer to the gun's electrodes. The closed gun has the position “0 mm”.

Zero position — During positioning, every specification is in relation to the zero position. This value changes only during travel calibration, so that the opening angle of the gun arms remains constant during positioning.

Note
- Carry out a travel calibration with new electrode caps only.

Reference position — The reference position refers to the opening width (sheet thickness) specified for welding and can change when homing is carried out (reason: burn-off and milling of the electrode caps; drift of the caps on the shafts, etc.).

The position of the drive with the gun arms closed which is determined during travel calibration is saved as the zero and reference position. The zero position remains constant until the travel calibration is performed again. The reference position can change each time homing is carried out.

After homing is carried out, the opening angle of the gun arms does not change during free positioning of the gun.

6.10.2 “Travel calibration” dialog box

1. Switch to the “Calibration” tab in the project window ➔ 6.9.3.
2. Open the “Travel calibration” dialog box using the corresponding button.

Travel calibration is performed automatically via this dialog box.

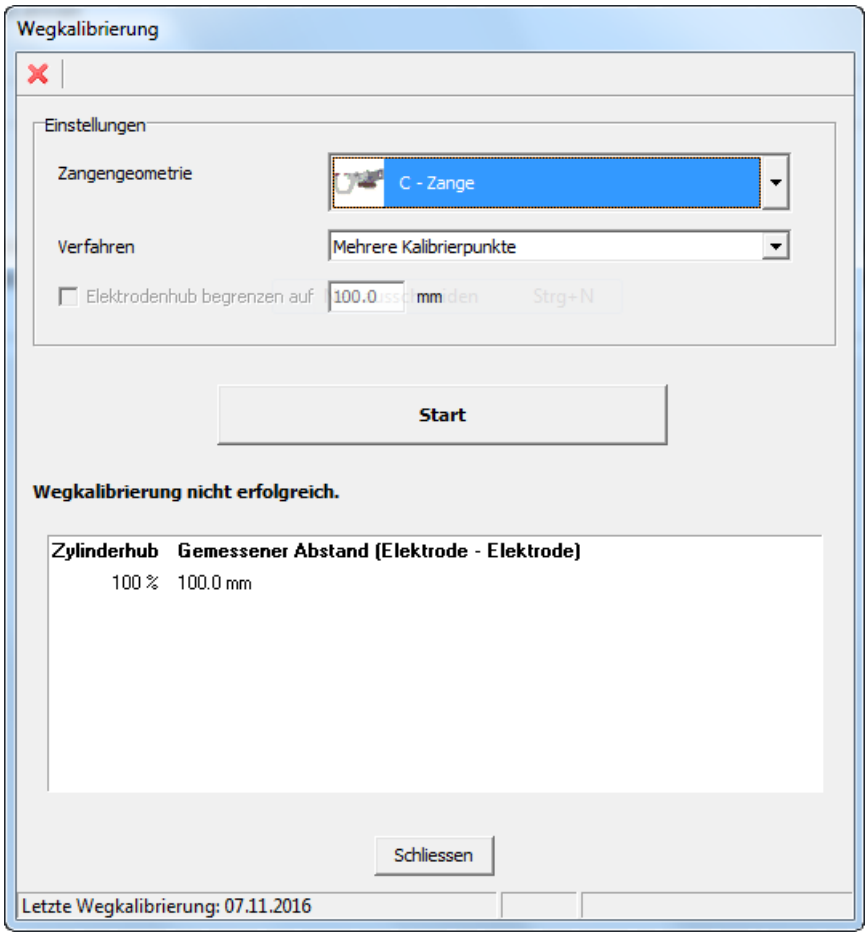


Figure 6-30 “Travel calibration” dialog box

Input field	Contents
Geometry	Gun geometry (C or X gun)
Method	– Method 1: One calibration point The gun ratio is determined after determining the zero position based on an approached opening width. – Method 2: Multiple calibration points The zero position, maximum opening width and 3 mid-positions are approached. Separate transmission ratios are calculated for each area.
Limit electrode stroke	Limits the electrode stroke to the set value. If a greater value is specified, error E006 is output. The limit can only be activated for Method 1.

Command	Button	Function
[Start]	–	Starts the travel calibration.
[Close]	–	Closes the dialog box.
–		“Error reset”, resets any errors present.

Performing and processing the travel calibration function



Warning
Risk of injury
Dynamic movements and high forces occur during travel calibration.

Step 1

- Start the travel calibration function with the “Start” button.

1. The welding gun arms move together until the set reference pressure is reached.
2. The pressure in the drive is reduced to release the stress on the welding gun arms.
3. The current position is saved as the zero and reference position.
4. If the “Multiple calibration points” method is selected, the main cylinder is retracted until it reaches a mechanical stop or it is fully retracted. This position is saved as the maximum opening width.

Step 2 (only with “Multiple calibration points” method)

- Measure the distance between the electrodes at full opening, 75 %, 50 % and 10 % cylinder stroke, and enter this value as the electrode stroke in the dialog box. The dialog box triggers the movement to the respective positions.
- Confirm each entry by pressing “OK”.

Gun	Method	Sequence
C	One calibration point	1. Determine zero position
	Multiple calibration points	1. Determine zero position 2. Arms are opened fully
X	One calibration point	1. Determine zero position 2. Open 10 %
	Multiple calibration points	1. Determine zero position 2. Arms are opened fully 3. Close to 75 % 4. Close to 50 % 5. Close to 10 %

Warning
Risk of injury
The drive is under pressure during the measurements.
The pressure in the main cylinder is locked in the cylinder by the additional pneumatic circuit.
- Do not remove any hoses or cables during the measuring process.

The friction detection factor is reset during travel calibration – the friction factor is determined again.

6.11 Force calibration

The force calibration is used to determine the ratio of transmission of cylinder force to electrode force. This is done by measuring the pliers forces corresponding to two specified cylinder forces.

In the controller CPX-WELD-2G-SA, the force setpoints are converted into cylinder forces based on the force calibration.

- Carry out the force calibration in the force range in which the welding gun is operated.

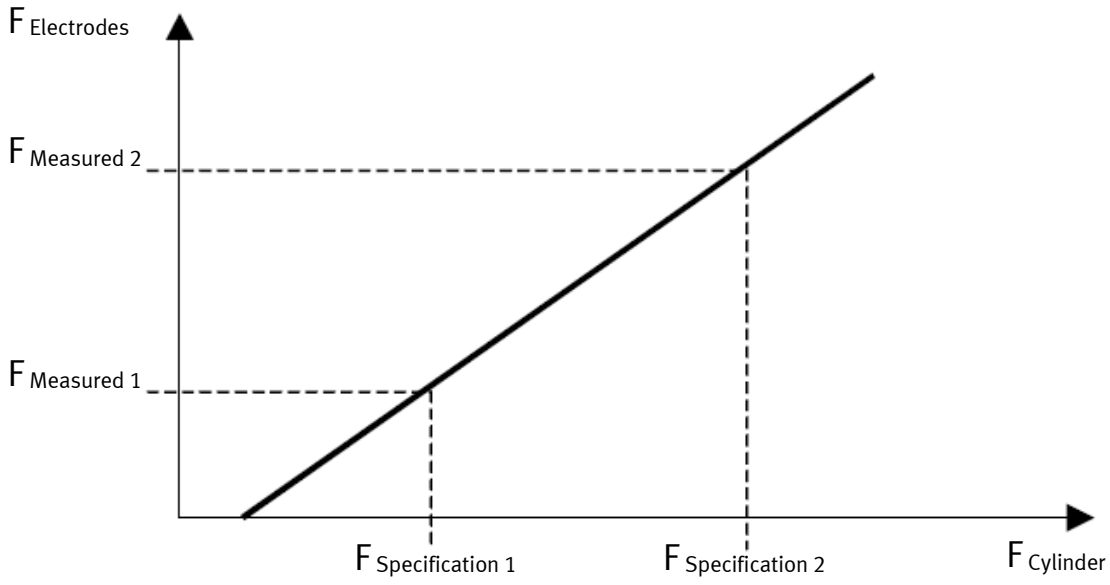
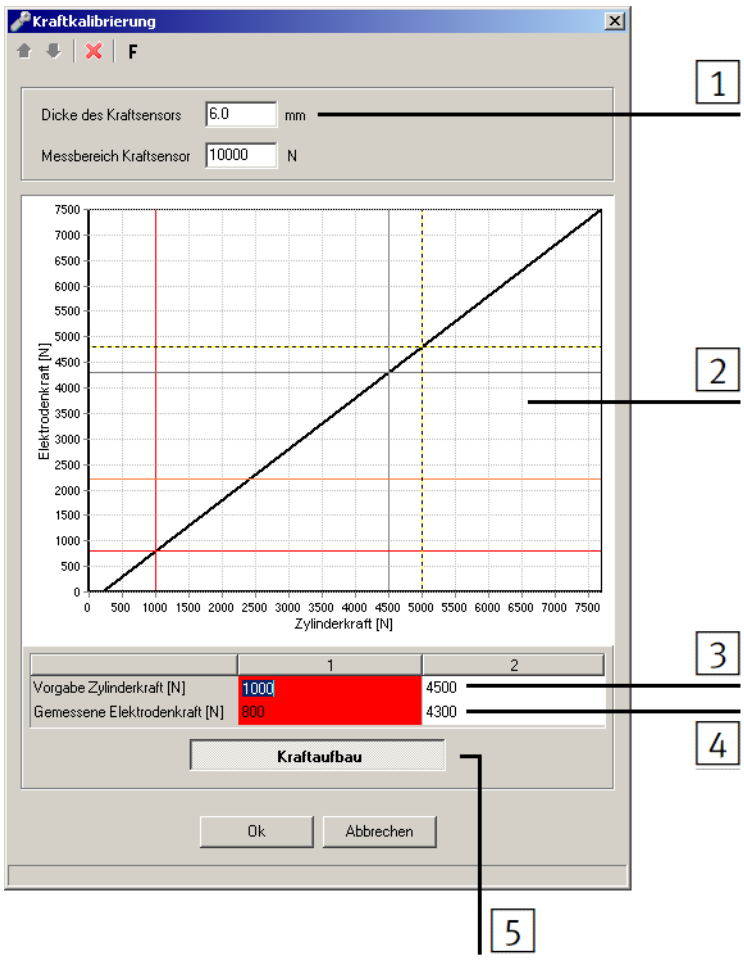


Figure 6-31 Force calibration diagram

6.11.1 “Force calibration” dialog window

1. Switch to the “Calibration” tab in the project window ➔ 6.9.3.
2. Open the “Force calibration” dialog window by using the corresponding button.

Force calibration is performed via this dialog window.



1 Set value to which the electrodes are to be closed for force build-up (the thickness of the force sensor in use).

2 Representation of the characteristic curve for electrode force – cylinder force

3 Set values for the force curve

4 Measured values for the force curve

5 Button for activating force build-up

Figure 6-32 “Force calibration” dialog window

Warning
Risk of injury
Dynamic movements and high forces occur during force calibration.

Command	Button	Function
[Open gun arms]		For manual movement of the electrodes.
[Close gun arms]		For manual movement of the electrodes.
[Error reset]		Resets an error.
[External force sensor]	F	Automatically applies the values of an external force sensor.
[Force build-up]	–	Positioning of the welding gun onto the force sensor and adjustment of the specified cylinder force.
[OK]	–	Applies the force calibration data to the project. When the online mode is active, the calibration data determined is automatically saved in the controller.
[Cancel]	–	Cancels the force calibration data. The data is discarded; it is not applied to the project, and calibration data is not transmitted to the controller.

Performance and process of force calibration

1. Enter the thickness of the force sensor you are using [1].
2. In column “1” [3], activate value pair 1 and enter the cylinder force.
3. Position the force sensor and then click “Force build-up” [5].
   The electrodes are positioned onto the force sensor and the welding force is built up.
4. Enter the measured force value in “Measured electrode force 1” [4].
5. Click “Force build-up” again – the gun opens.
6. In column “2”, activate value pair 2 and repeat steps 3 to 5 accordingly.
7. Click “OK” to save the values to the controller.

6.12 Control of the welding gun

WinSPAS can be used to carry out all the functions which can also be carried out by the fieldbus.

6.12.1 Functions of the “Controller” panel

Warning

Possible risk of injury and material damage.

When switching master control (PC controller) to the fieldbus, the values set through the fieldbus apply immediately (e.g. for position). As a result, the main cylinder and the compensating cylinder can move with great force and at high speed.

When operated via the “Controller” panel, the cylinders perform movements with high dynamics and force.

- Make sure that nobody can enter the positioning range of moving parts.

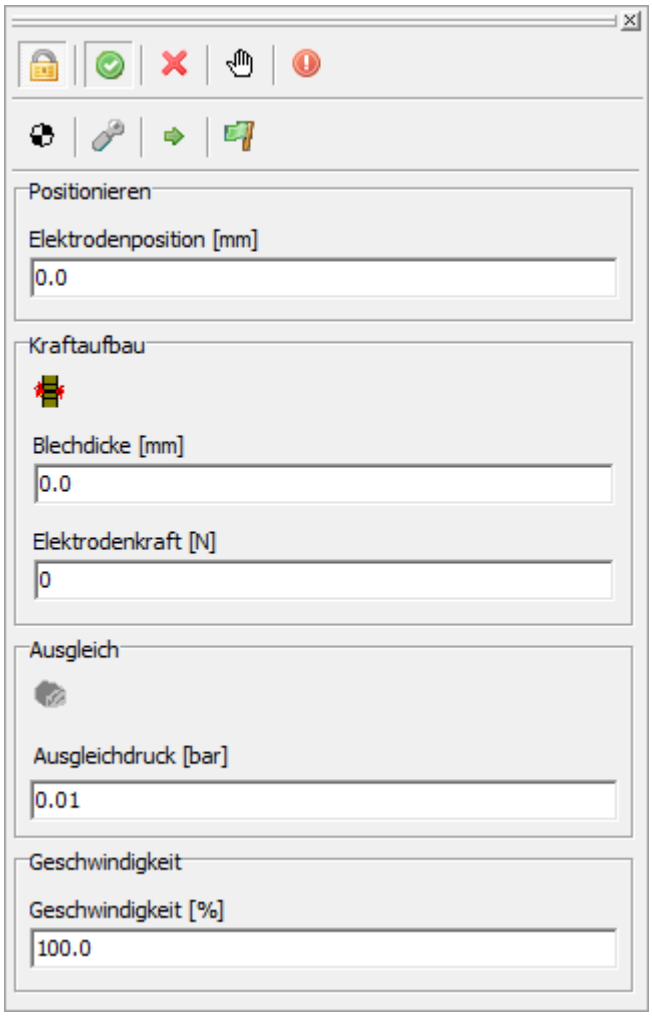
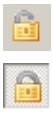


Figure 6-33 “Controller” panel

Command	Button	Function
PC control		Clicking this button switches master control to the PC (indicated by the depressed button). Activation of the PC controller interrupts the current commands issued by the external controller. Clicking the pressed button again switches control of the drive back to the fieldbus master.
[Enable]		This button is used to enable and re-enable positioning for the system. Enable = 1 is a prerequisite for any positioning of the drive.
[Error reset]		Any error present on the CPX-WELD-2G-SA can be reset by clicking this button. If an error cannot be reset, the cause of the error still exists.
[Manual / Auto]		When the manual mode is selected (button not depressed), the welding gun moves at a reduced speed. When the automatic mode is selected (button depressed), the welding gun moves at maximum speed.
[Stop]		Clicking this button disables drive positioning. The valves move into their mid-position immediately and the current command is discarded. Positioning must be re-enabled by clicking [Enable] again.
[Homing]		This button is used to start the homing run. The drive moves the gun slowly together and determines the closed position of the arms. Clicking this button again stops the homing run. The drive moves to the specified setpoint position.
[Force calibration]		Starts the force calibration procedure via the fieldbus. • Before pressing this button, connect the external force sensor to interface AI 4 on the controller. The controller positions the electrodes onto the force sensor in two steps and measures the electrode forces for the cylinder forces set under force calibration.

Command	Button	Function
[Output 1]		Controls the available output 1.
[Air supply shutdown]		Activates/deactivates the air supply shutdown function. When air supply shutdown is activated (red flag) the air supply in the control block is cut off. The air in the main drive is locked in by the additional pneumatic circuit. Some errors are suppressed in this state ➔ A.2 Error numbers).

If the “PC controller” is still active when exiting the software package WinSPAS, a warning message will be displayed.

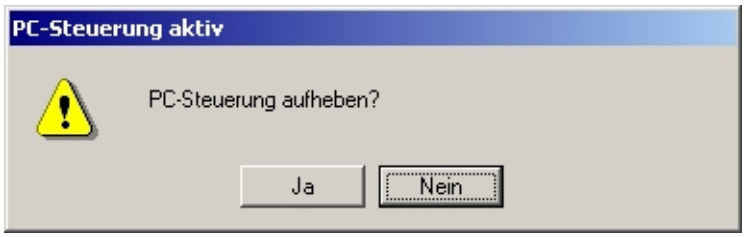


Figure 6-34 Warning message “PC controller active”

Warning

Possible risk of injury and material damage.

When switching master control (PC controller) to the fieldbus, the values set through the fieldbus apply immediately (e.g. for position). As a result, the main cylinder and the compensating cylinder can move with great force and at high speed.

When operated via the “Controller” panel, the cylinders perform movements with high dynamics and force.

- Make sure that nobody can enter the positioning range of moving parts.

Positioning

Electrode position

A position setpoint can be entered via the input field. When confirming your entry with the RETURN button, this value is sent to the controller.
When “Force build-up” or “Homing” is completed, the drive travels to this position.

Force build-up

 [Force build-up]	This button is used to start force build-up. The relevant setpoint values can be set via this panel: – Sheet thickness (= opening width) – Electrode force (= welding force) The drive is positioned to the opening width. When this position is reached, the controller switches to force control. Force build-up remains active as long as the [Force build-up] button is indented. The force setpoint can be changed at any time, even when force build-up is active.

Compensation

 [Compensation]	This button can be used to simulate the overall behaviour (movement of the main drive and compensation) when welding. While the button is indented, the pressure value is switched over automatically.
Compensating pressure	A compensating pressure setpoint can be entered via the input field. When confirming your entry with the RETURN button, this value is sent to the controller. • Right-click the [Compensation] button and select “Set values”. The two compensating pressures can be set in the following dialog box.

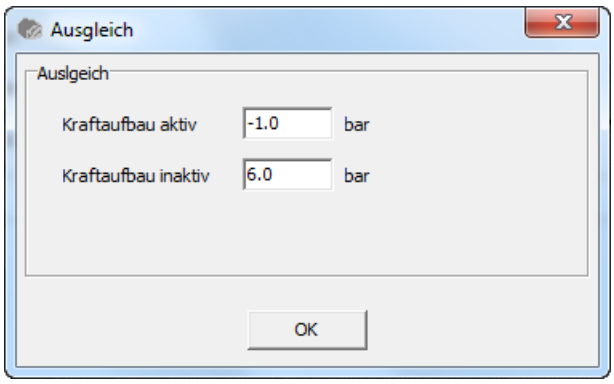


Figure 6-35 “Compensation” dialog box

The differential pressure in the compensating cylinder can be entered via the following input fields.

Input field	Contents
Force build-up active	Pressure setpoint of the compensating cylinder during force build-up.
Force build-up inactive	Pressure setpoint of the compensating cylinder during positioning.

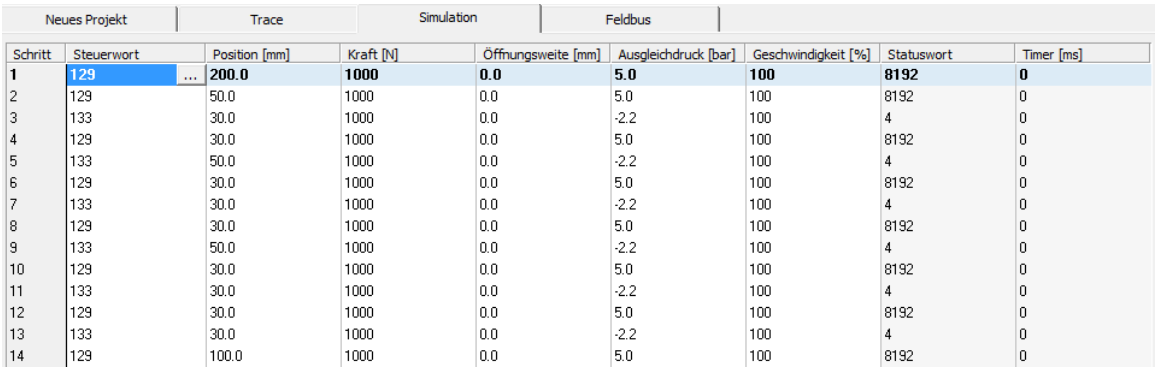
The pressure setpoint for “force build-up inactive” must be greater than the pressure setpoint for “force build-up active”. This ensures that the welding gun is pressed into a fixed stop during the inactive phase.

Speed

Speed	With this value, you can specify the speed of the main drive via the fieldbus. The speed can also be permanently reduced in the project data (necessary for large guns to limit the torque) ➔ Chapter 6.9.2, section “Speed and acceleration”. If 50 % has been parameterised here, for example, the fieldbus speed parameter only takes effect when a value of less than 50 % is specified.

6.13 Simulation

You can use the “Simulation” tab to program a procedure for testing the welding gun system.



Schritt	Steuerwort	Position [mm]	Kraft [N]	Öffnungsweite [mm]	Ausgleichdruck [bar]	Geschwindigkeit [%]	Statuswort	Timer [ms]
1	**129** ...	**200.0**	**1000**	**0.0**	**5.0**	**100**	**8192**	**0**
2	129	50.0	1000	0.0	5.0	100	8192	0
3	133	30.0	1000	0.0	-2.2	100	4	0
4	129	30.0	1000	0.0	5.0	100	8192	0
5	133	50.0	1000	0.0	-2.2	100	4	0
6	129	30.0	1000	0.0	5.0	100	8192	0
7	133	30.0	1000	0.0	-2.2	100	4	0
8	129	30.0	1000	0.0	5.0	100	8192	0
9	133	50.0	1000	0.0	-2.2	100	4	0
10	129	30.0	1000	0.0	5.0	100	8192	0
11	133	30.0	1000	0.0	-2.2	100	4	0
12	129	30.0	1000	0.0	5.0	100	8192	0
13	133	30.0	1000	0.0	-2.2	100	4	0
14	129	100.0	1000	0.0	5.0	100	8192	0

Figure 6-36 “Compensation” dialog box

One row in the table represents one simulation step. The simulation switches to the next step once the desired status word has been received and the time under “Timer” has elapsed.

Setting the control and status words

... Click this button to set the control or status word. A dialog box appears for setting the control or status word.

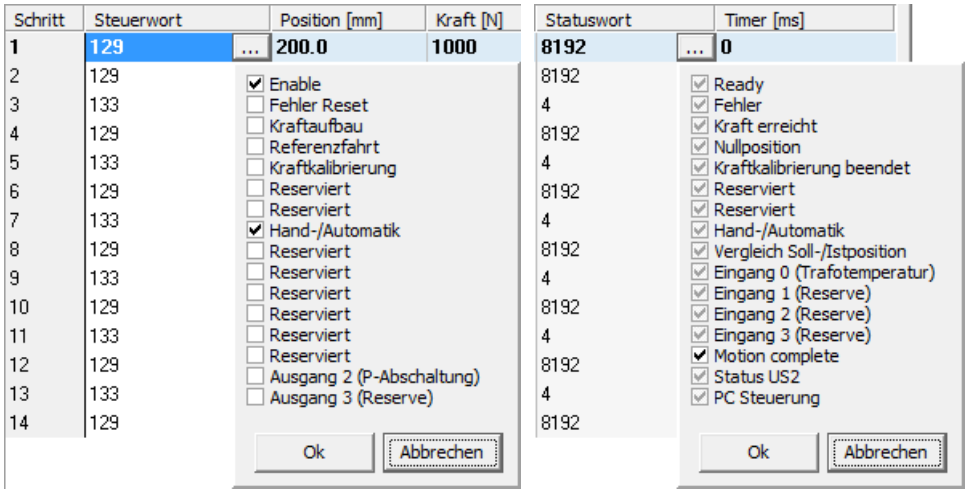
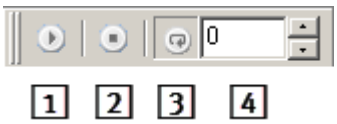


Figure 6-37 Dialog box for control and status words

Check box	Significance
☑	The corresponding bit in the status word must be set.
☐	The corresponding bit in the status word must be reset.
☑ (grey)	The state of the corresponding bit in the status word has no significance.

Controlling the simulation

The simulation is controlled via the toolbar:



1 Start simulation

2 Stop simulation

3 Repeat continuously, in an endless loop

4 Number of repetitions

Figure 6-38 Dialog box for control and status words

The PC controller must be activated to start the simulation.
You can continue to change the project data during the simulation run.
You can also start a trace recording at any time.
The simulation stops when the trace data is uploaded.

6.14 Firmware update

You can update the firmware for the controller by using the WinSPAS program.

Note

Damage to the product

- The logic voltage supply US1 must not be switched off during the firmware update.

The firmware update takes several minutes.

- Before starting it, switch off the compressed air supply.

- Open the “Firmware update” dialog box by accessing the [Options] menu and clicking [Firmware], then [Firmware update].

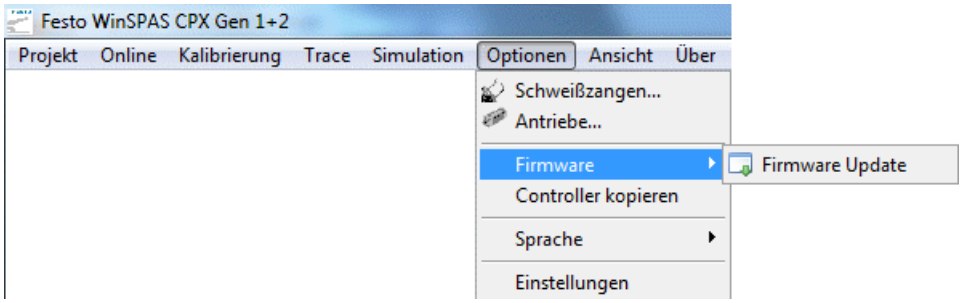


Figure 6-39 Menu command [Firmware] [Firmware update]

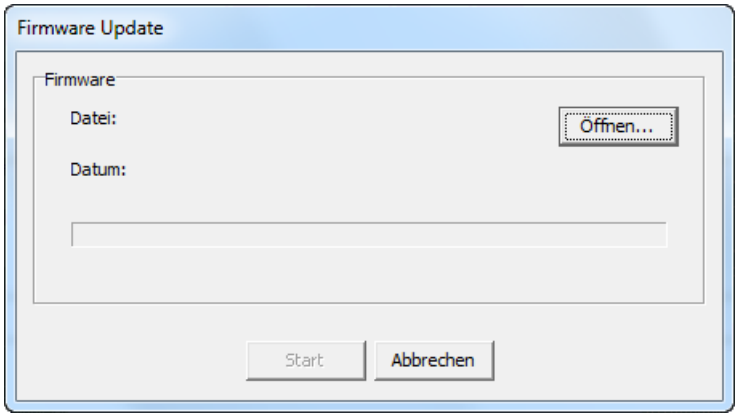


Figure 6-40 “Firmware Update” dialog box

1. Click “Open” to select a firmware file.
2. Click “Start” to transfer the firmware to the controller.

7. Control via fieldbus

7.1 Fieldbus protocol

#	Inputs			Outputs		
0.00	Control word	Enable		Control word	Ready	Gun ready for operation (no error, US2 present, closed-loop controllers active)
0.01		Error reset			Error	Error occurred
0.02		Force build-up	Close gun arms and build up force		Force reached	[1] Welding force reached
0.03		Homing	Execute a homing run		Zero position	[1] Homing completed
0.04		Force calibration	[1] Start the process for force calibration		Force calibration finished	[1] Force scaling finished
0.05		Reserved			Reserved	
0.06		Reserved			Reserved	
0.07		Manual/automatic	[0] Manual [1] Automatic		Manual/automatic	[0] Manual [1] Automatic
0.08		Reserved			Setpoint/actual position comparison (Drift detection)	When cyl. locked: [0] Position OK [1] Position outside tolerance range
0.09		Reserved			Input 0	
0.10		Reserved			Input 1	
0.11		Reserved			Input 2	
0.12		Reserved			Input 3	
0.13		Reserved			Motion complete	[1] Setpoint position reached
0.14		Air supply shutdown	[0] P-supply shutdown inactive [1] P-supply shutdown active		Status US2	[0] US2 = 0 V [1] US2 > 17 V
0.15		Output 1	Available output		PC controlled	

#	Inputs			Outputs		
1.00	Setpoint position	Bit 0	0.1 mm / bit INT	Current position	Bit 0	0.1 mm / bit INT
1.01		Bit 1			Bit 1	
1.02		Bit 2			Bit 2	
1.03		Bit 3			Bit 3	
1.04		Bit 4			Bit 4	
1.05		Bit 5			Bit 5	
1.06		Bit 6			Bit 6	
1.07		Bit 7			Bit 7	
1.08		Bit 8			Bit 8	
1.09		Bit 9			Bit 9	
1.10		Bit 10			Bit 10	
1.11		Bit 11			Bit 11	
1.12		Bit 12			Bit 12	
1.13		Bit 13			Bit 13	
1.14		Bit 14			Bit 14	
1.15		Prefix			Prefix	

#	Inputs			Outputs		
2.00	Setpoint electrode force	Bit 0	1N/Bit UINT	Current electrode force	Bit 0	1N/Bit UINT
2.01		Bit 1			Bit 1	
2.02		Bit 2			Bit 2	
2.03		Bit 3			Bit 3	
2.04		Bit 4			Bit 4	
2.05		Bit 5			Bit 5	
2.06		Bit 6			Bit 6	
2.07		Bit 7			Bit 7	
2.08		Bit 8			Bit 8	
2.09		Bit 9			Bit 9	
2.10		Bit 10			Bit 10	
2.11		Bit 11			Bit 11	
2.12		Bit 12			Bit 12	
2.13		Bit 13			Bit 13	
2.14		Bit 14			Bit 14	
2.15		Bit 15			Bit 15	

#	Inputs			Outputs		
3.00	Compensating pressure	Bit 0	305 µbar / bit UINT		Reserved	
3.01		Bit 1			Reserved	
3.02		Bit 2			Reserved	
3.03		Bit 3			Reserved	
3.04		Bit 4			Reserved	
3.05		Bit 5			Reserved	
3.06		Bit 6			Reserved	
3.07		Bit 7			Reserved	
3.08		Bit 8		Error number	Bit 0	Error number UINT
3.09		Bit 9			Bit 1	
3.10		Bit 10			Bit 2	
3.11		Bit 11			Bit 3	
3.12		Bit 12			Bit 4	
3.13		Bit 13			Bit 5	
3.14		Bit 14			Bit 6	
3.15		Bit 15			Bit 7	

#	Inputs			Outputs		
4.00	Sheet metal thickness	Bit 0	0.1 mm / bit UINT	Cap burn-off	Bit 0	0.1 mm / bit INT
4.01		Bit 1			Bit 1	
4.02		Bit 2			Bit 2	
4.03		Bit 3			Bit 3	
4.04		Bit 4			Bit 4	
4.05		Bit 5			Bit 5	
4.06		Bit 6			Bit 6	
4.07		Bit 7			Bit 7	
4.08		Bit 8			Bit 8	
4.09		Bit 9			Bit 9	
4.10		Bit 10			Bit 10	
4.11		Bit 11			Bit 11	
4.12		Bit 12			Bit 12	
4.13		Bit 13			Bit 13	
4.14		Bit 14			Bit 14	
4.15		Bit 15			Bit 15	

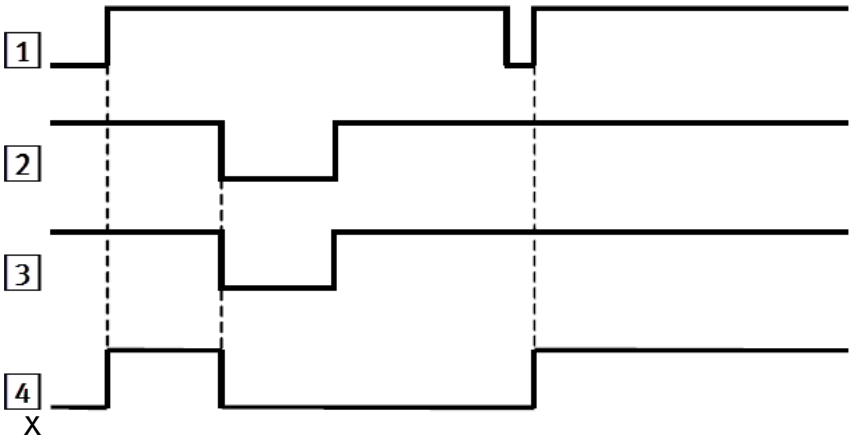
#	Inputs			Outputs		
5.00	Speed in automatic	Bit 0	UCHAR 255 = Full speed Speed, if [E0.07] automatic = 1		Reserved	
5.01		Bit 1			Reserved	
5.02		Bit 2			Reserved	
5.03		Bit 3			Reserved	
5.04		Bit 4			Reserved	
5.05		Bit 5			Reserved	
5.06		Bit 6			Reserved	
5.07		Bit 7			Reserved	
5.08		Reserved			Reserved	
5.09		Reserved			Reserved	
5.10		Reserved			Reserved	
5.11		Reserved			Reserved	
5.12		Reserved			Reserved	
5.13		Reserved			Reserved	
5.14		Reserved			Reserved	
5.15		Reserved			Reserved	

7.2 Programming the higher-order controller

7.2.1 General information

For all movements of the gun, the bit [0.00] (Enable) must be set and the load voltage supply US2 must be present. Bit [0.07] (Manual/Automatic) should be set in normal operation.

7.2.2 Enable

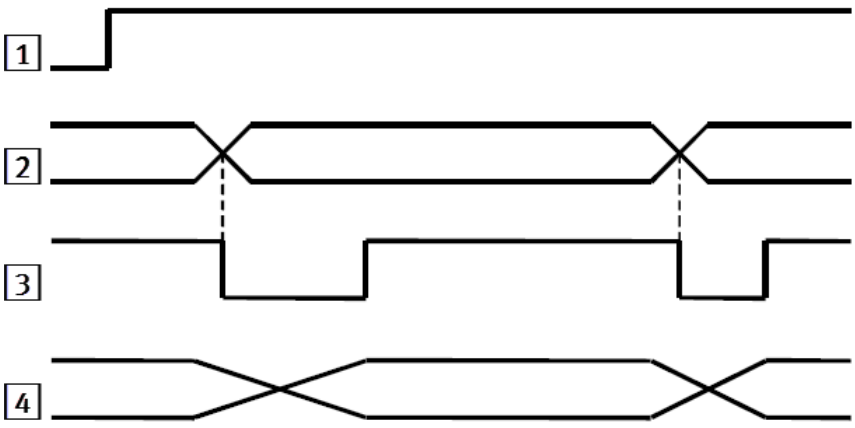


1 [E0.0] Enable
2 US2
3 [A0.14] Status US2
4 Positioning enabled

Figure 7-1 Enable

During the first movement of the welding gun after the device is switched on, the direction of motion is monitored. If the correct direction of motion is reported, this monitoring is switched off.

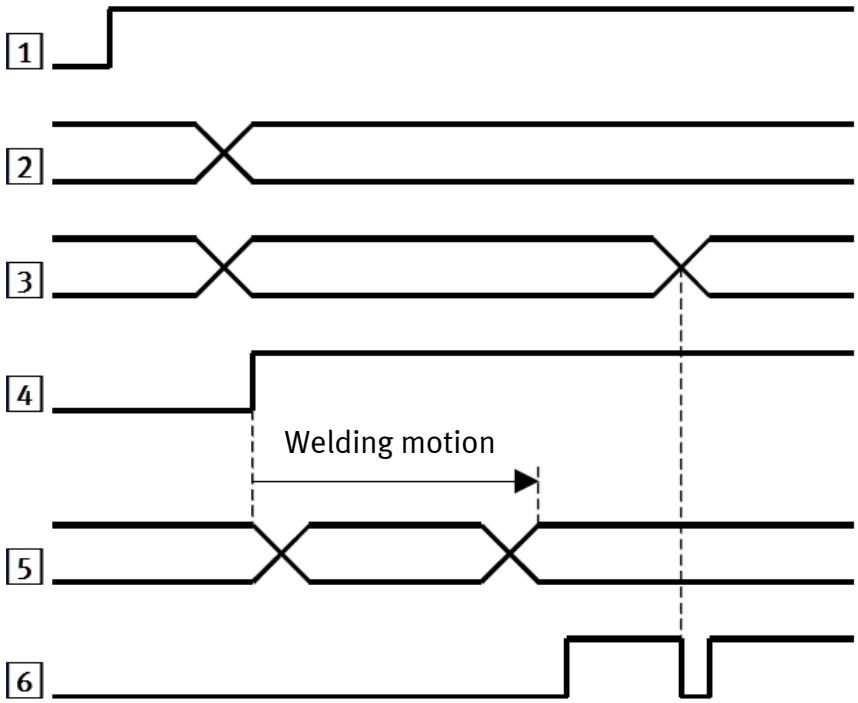
7.2.3 Positioning



1 [E0.00] Enable
2 [E1.00 ... 1.15] Setpoint position
3 [A0.04] Motion Complete
4 [A1.00 ... 1.15] Actual position

Figure 7-2 Enable

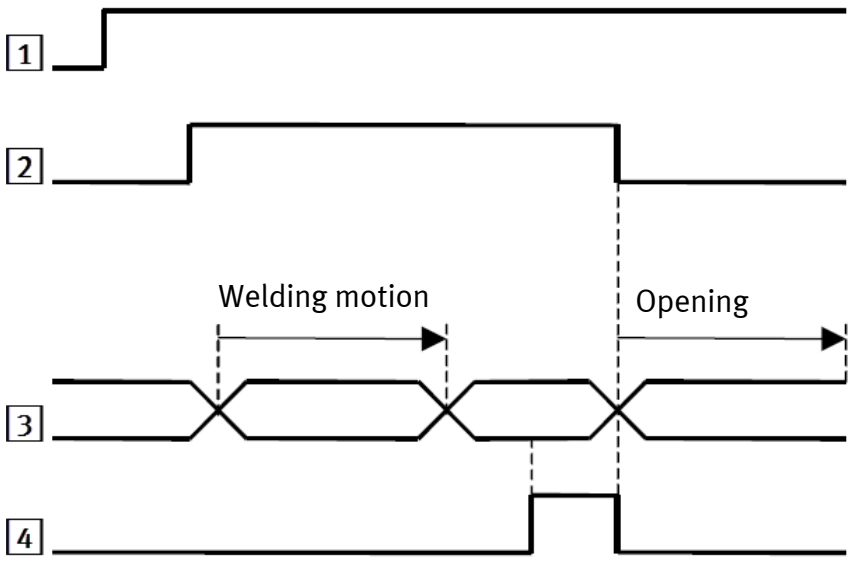
7.2.4 Welding



1 [E0.00] Enable
2 [E4.00 ... 4.15] Opening width
3 [E2.0 ... 2.15] Welding force
4 [E0.02] Force build-up
5 [A1.00 ... 1.15] Actual position
6 [A0.02] Force reached

Figure 7-3 Welding

7.2.5 Homing



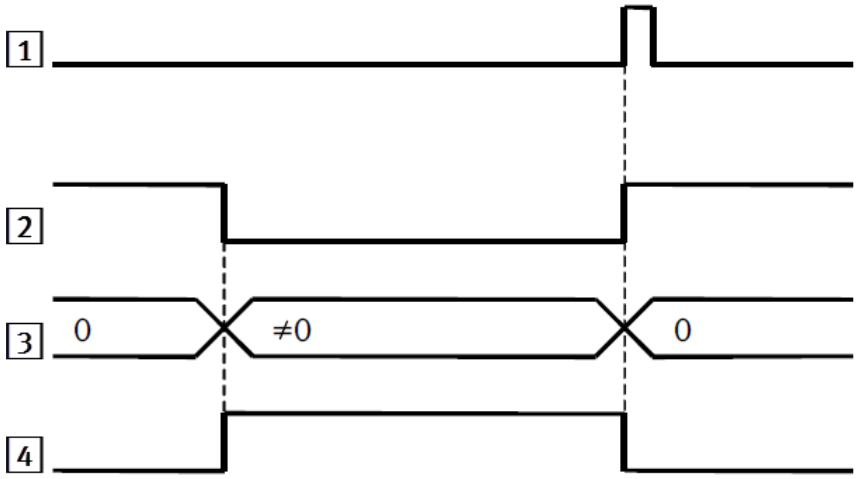
1 [E0.00] Enable
2 [E0.03] Homing
3 [A1.00 ... 1.15] Actual position
4 [A0.03] Zero position

Figure 7-4 Homing

During the homing procedure the gun moves at a reduced speed.

- Carry out homing from a small opening width.
- Always start homing from the same position.

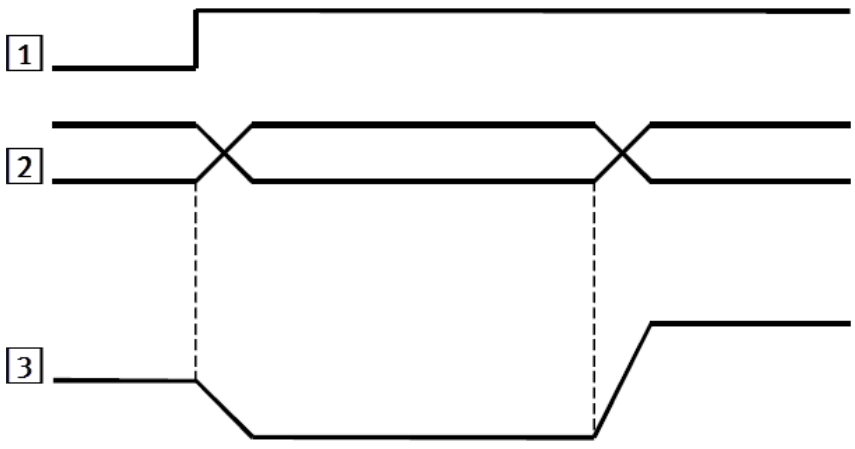
7.2.6 Error handling



1 [E0.01] Error reset
2 [A0.00] Ready
3 [A3.8 ... 3.15] Error number
4 [A0.01] Error

Figure 7-5 Error handling

7.2.7 Compensation



1 [E0.00] Enable

2 [E3.00 ... 3.15]
Setpoint value, compensating pressure

3 Pressure in compensating cylinder

Figure 7-6 Compensation

7.3 Diagnostics

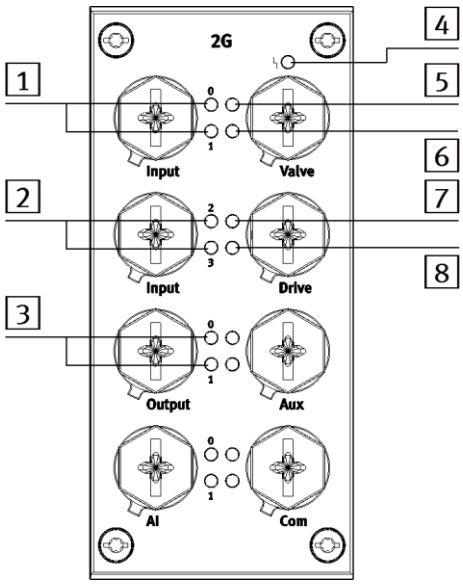
7.3.1 Diagnostics via fieldbus

The controller's protocols contain the following bits, which indicate the status of the controller:

- [A0.00] Ready
- [A0.01] Error
- [A3.08 ... 3.15] Error number

7.3.2 Diagnostics on the CPX-WELD-2G-SA

Significance of the LEDs on the controller:



1 Input: Input 0, Input 1
2 Input: Input 2, Input 3
3 Output: Output 0, Output 1
4 Error
5 Positioning enabled
6 Status of load voltage supply US2
7 Position control
8 Force control

Figure 7-7 LEDs on CPX-WELD-2G-SA

7.3.3 Diagnostics via WinSPAS

WinSPAS supports diagnostics with:
- Display of setpoints and actual values
- Display of fieldbus data
- Trace function

“Setpoint/actual values” panel

When the online mode is selected, the “Setpoint/actual values” panel displays the cyclical data in real time.

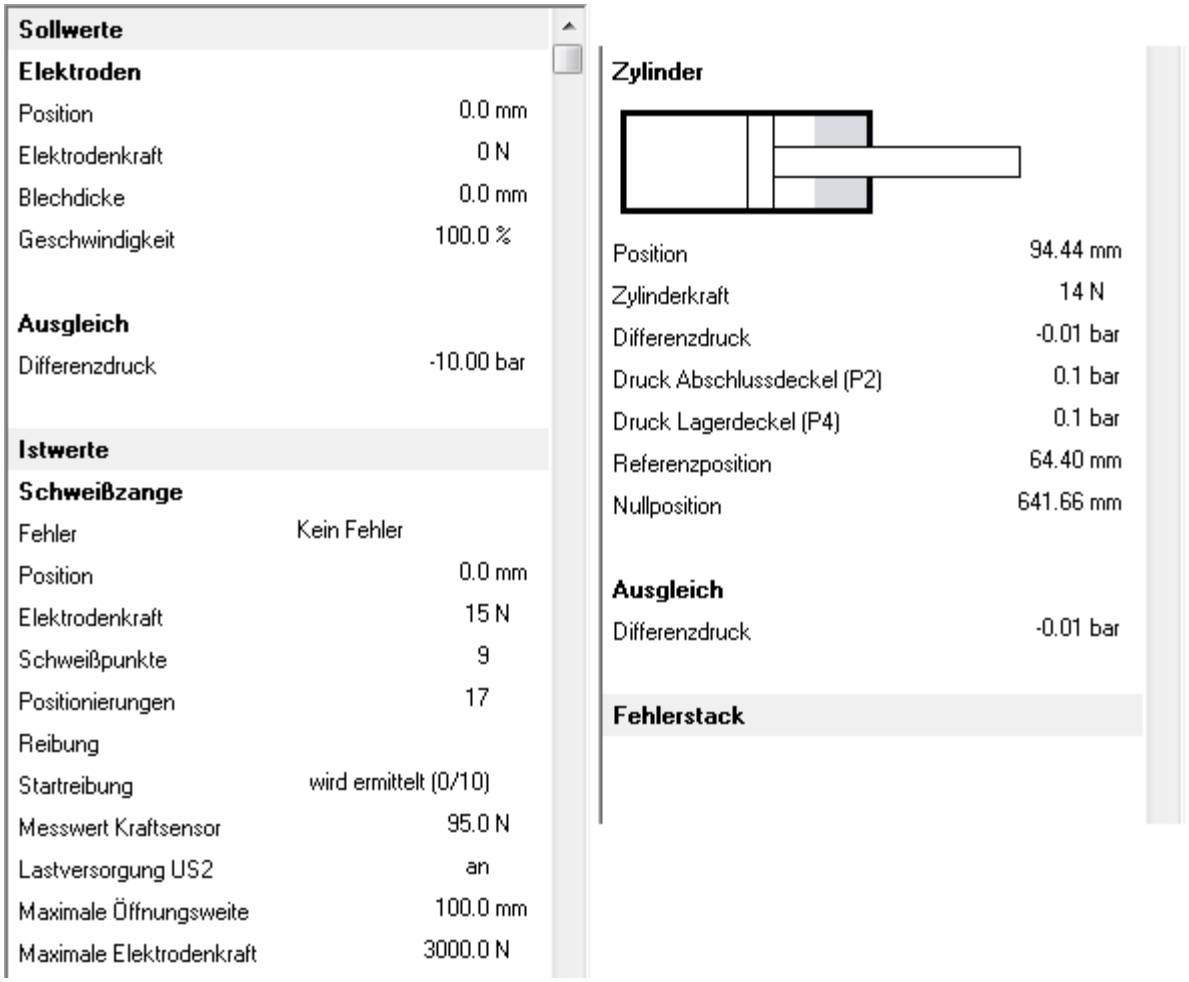


Figure 7-8 “Setpoint/actual values” panel

A. Technical data

A.1 General information

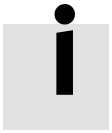
General information		Value
Nominal pressure	[bar]	5 ... 12
Operating pressure	[bar]	≤ 12
Permissible overpressure (t ≤ 1 sec, no damage)	[bar]	16
Operating medium at main port 1 of the control block		Compressed air to ISO 8573-1:2010 [6:4:2] [1)]
Operating voltage		
Logic voltage supply US1	[V DC]	24 ± 10 %
Load voltage supply US2	[V DC]	24 ± 10 %
Rated operating current (load and logic voltage supply)	[A]	≤ 4
Load current for digital outputs	[mA]	≤ 250
Ambient conditions		
Ambient temperature °C	[°C]	0 ... 50
Storage temperature	[°C]	–10 ... +60
Temperature of medium	[°C]	5 ... +40
Relative humidity (non-condensing)	[%]	≤ 90
Degree of protection (DIN EN 60529)		IP 65 [2)]
Electromagnetic compatibility		
Emitted interference		to EN 61000-6-4 [3)]
Immunity to interference		to EN 61000-6-2
Shock resistance (IEC68-2-27) severity level 2 [4)]		30 g
Vibration resistance (IEC68-2-6) severity level 2 [4)]		5 g

1) A 5 µm filter is integrated in the manifold block.
2) Note: All electrical connections must be capped. The protective transport cap does not constitute IP protection.
3) The component is intended for industrial use.
4) Note: Assemble the welding gun controller and control block using rubber buffers.

Table A.1 Technical data – General

A.2 Error numbers

A pending error can be deleted by means of a positive “Acknowledge error – edge” (error reset).



The cause of the error must be eliminated before it can be deleted with “Error reset”.

No.	Error	Detection			
		CPX	MPYE	MPYD	Cylinder
1	– (not assigned)				
2	– (not assigned)				
3	Compressed air supply	X			
4	Max. cylinder force	X			
5	Memory error	X			
6	Position specification	X			
7	– (not assigned)				
8	Force specification	X			
9	Position timeout	X			
10	Force timeout	X			
11	Direction of motion	X			
12	Timeout compensation	X			
13	Parameter	X			
14	– (not assigned)				
15	Force calibration	X			
16	Electrode monitoring	X			
17	Friction too low	X			
18	Friction too high	X			
19	Sheet thickness	X			
20	Short circuit at output 0	X			
21	Short circuit at output 1	X			
22	Short circuit block control valve (output 2)	X			

No.	Error	Detection			
		CPX	MPYE	MPYD	Cylinder
23	Short circuit air saving control valve (output 3)	X			
24	– (not assigned)				
25	– (not assigned)				
26	Parameterisation	X			
27	Welding position	X			
28	Unintentional movement	X			
29	Drift	X			
30	Blocking function of main cylinder				X
31	– (not assigned)				
32	– (not assigned)				
33	Drive CAN communication	X			
34	Drive position sensor				X
35	Drive pressure sensor LD (P4)				X
36	Drive pressure sensor AD (P2)				X
37	Drive voltage				X
38	Drive diagnostic memory				X
39	Drive parameter memory				X
40	– (not assigned)				
41	– (not assigned)				
42	– (not assigned)				
43	Drive valve CAN communication	X			
44	Drive valve clamping		X		
45	Drive valve temperature		X		
46	Drive valve voltage		X		
47	– (not assigned)				
48	– (not assigned)				
49	Drive valve comparison		X		
50	– (not assigned)				

No.	Error	Detection			
		CPX	MPYE	MPYD	Cylinder
51	– (not assigned)				
52	– (not assigned)				
53	Compensating valve CAN communication	X			
54	Compensating valve clamping			X	
55	Compensating valve temperature			X	
56	Compensating valve voltage			X	
57	– (not assigned)				
58	– (not assigned)				
59	Compensating valve comparison			X	
60	– (not assigned)				
61	– (not assigned)				
62	– (not assigned)				
63	Configuration error	X			
64	– (not assigned)				

Table A.2 Technical data – Error numbers

B. Transport conditions

The following transport conditions must be observed to ensure that the controller is not damaged during further or return transport:

- Always use the original packing for transport.
- The ambient conditions for the device must also be observed during transport ➔ A Technical data.

C. Notes on disposal



- Observe the local regulations regarding waste disposal and the environment.

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